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Music During Study and Work: A Scoping Review of Cognitive Performance, Activity Demand, and Individual Differences

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Abstract

Listening to music while studying or working has become a common feature of contemporary life, yet evidence on whether music supports or disrupts cognitive performance remains inconsistent. This thesis presents a scoping review of empirical research on how music affects memory, learning, motivation, and concentration during study and work-related cognitive tasks. It also examines how these effects are shaped by task demand, musical characteristics, and individual differences. Guided by the methodological framework of Arksey and O'Malley and reported in accordance with PRISMA-ScR, the review searched seven databases and charted 30 empirical studies published between 2010 and 2026. The evidence was interpreted through an activity-demand framework grounded in cognitive load theory and the arousal–mood hypothesis.

Three main findings emerged. First, the direction of music's effect varied with the cognitive demand of the activity. Music tended to support performance on routine, low-load activities, produced more conditional effects on familiar but effortful activities, and was more likely to impair performance on novel, high-load activities such as reading comprehension. Second, impairment appeared to be linked especially to lyrical content rather than to music itself, with semantically rich lyrics, particularly in the listener's native language, being most disruptive to verbal tasks. Instrumental music was often neutral and, in some cases, beneficial. Third, individual differences in working-memory capacity, personality, music preference, and listening habit shaped both the strength and direction of these effects. Learners' subjective sense of benefit did not always correspond with measured performance.

The review concludes that music should be understood as a context-dependent influence on learning rather than as a general cognitive enhancer. It also identifies motivation, tempo, and realistic workplace activities as priorities for future research.

Keywords: background music, cognitive performance, scoping review, cognitive load, activity demand, reading comprehension

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Chapter 1: Introduction

1.1 Background and Rationale

Listening to music while studying or working has become a routine part of everyday life for many learners and workers. The widespread availability of streaming services and portable listening devices means that learners and workers can often accompany many activities with music of their own choosing. Surveys of student behaviour indicate that many learners listen to music during at least part of their study time, often because they believe it improves their focus, motivation, or mood (Goltz & Sadakata, 2021). Whether this belief is justified, and whether music supports or undermines the cognitive work it accompanies, is therefore not only a laboratory question but also an issue of everyday significance for students, educators, and employers. It is also a question about the conditions under which people learn and work, which makes it relevant to educational research.

Research on this question dates back almost a century. Early investigations examined whether music in factories could sustain output and morale during repetitive work (Wyatt & Langdon, 1937; Uhrbrock, 1961). A separate line of study examined music in classrooms and learning, and a third, prompted by the widely publicised and later contested "Mozart effect" (Rauscher et al., 1993), asked whether simply listening to music could raise cognitive ability itself. What links these traditions is the persistence of inconsistent findings. Across this body of research, studies have reported beneficial, harmful, and neutral effects of music on a range of cognitive outcomes. It is this inconsistency that the present review takes as its starting point.

Emerging evidence suggests that the effects of music on cognitive performance may depend on the conditions under which it is used rather than following a single, consistent pattern. Studies have reported that music may support performance during some routine or relatively undemanding activities while interfering with more cognitively demanding tasks such as reading comprehension and verbal learning, although the findings remain mixed. This variation suggests that differences in the cognitive demands of the activity, the characteristics of the music, and the characteristics of the listener may all contribute to the outcomes observed. Accordingly, rather than asking whether music is generally beneficial or detrimental for cognitive performance, this thesis examines the conditions under which different effects occur during study and work-related cognitive tasks. To address this question, the review draws on established theories of arousal, attention, working memory, and individual differences, which are developed later in this thesis and brought together through an activity-demand framework for organising and interpreting the evidence. Understanding these conditions is important not only for explaining the existing literature but also for informing how learners and educators make decisions about the use of music during cognitively demanding activities.

1.2 Research Question and Aim

The central research question of the thesis is:
How does music affect memory, learning, motivation, and concentration during study and work-related cognitive tasks, and under what conditions do these effects occur?
This question is examined through three more specific sub-questions:
1. How does the effect of music differ according to the type and cognitive demand of the activity being performed?
2. How do the characteristics of the music influence its effect?
3. How do differences between individuals influence its effect?

Rather than attempting to determine whether music is universally beneficial or detrimental, this review seeks to identify the conditions under which different effects are most likely to occur.

The aim is to map the empirical evidence relating to these questions across educational and occupational settings, across non-clinical populations of different ages, and across the range of musical conditions examined in the literature, and to organise that evidence using the activity-demand framework. A scoping review was chosen for this purpose. Because the research question is broad and multi-part, and the literature it addresses is marked by varied methods and inconsistent findings, the appropriate first task is to map the breadth of the evidence, to clarify its key concepts, and to identify its gaps. These are the defining purposes of a scoping review (Arksey & O'Malley, 2005; Munn et al., 2018), rather than to calculate or synthesise a single overall effect, as would usually be expected in a more narrowly focused systematic review or meta-analysis. The methodology is described in full below.

1.3 Significance of the Review

The review aims to make three contributions. The first is educational. By mapping the evidence on music as a condition for studying and working, and interpreting it through a theoretically grounded framework, this work helps to organise an apparently contradictory literature and provides a basis for more careful, conditional guidance for learners and educators. It addresses a behaviour that learners engage in frequently but about which they often receive limited evidence-based advice.

The second contribution concerns scope. The review provides a transparent map of a literature that appears to have grown substantially in recent years, particularly since 2020, and that has less often been charted with educational and workplace evidence considered side by side. By considering evidence from different settings and types of cognitive activity, it may help identify patterns that a narrower review might overlook, especially when the impact of music is influenced more by the cognitive requirements of the task than by the environment in which it occurs.

This contribution can be located in relation to the existing review literature. Earlier reviews have approached the question in different ways: the meta-analysis by Kämpfe et al. (2011) pooled effects across studies and found the overall impact of background music on adults to be close to null; Cheah et al. (2022) reviewed how task, music, and population factors shape the effect; de la Mora Velasco et al. (2023) synthesised the evidence for learners specifically; and Mendes et al. (2021) reviewed the effect of music on attention, concluding that music tends to support attention largely through mood and motivation while noting that methodological diversity makes the findings difficult to generalise. These syntheses acknowledge task characteristics to differing degrees, but none adopts an explicit, theory-based account of activity demand as its organising principle for mapping the evidence, unifies educational and occupational settings within a single framework, or treats individual differences as central rather than incidental.

This review aims to contribute to the existing literature in this specific area: not by recalculating a comprehensive effect, as the meta-analytic evidence indicates that such an approach would obscure more than it clarifies, but by illustrating how the influence of music varies based on the demands of the activity, the nature of the music, and the attributes of the listener.

The third contribution is to identify areas where the evidence remains limited, particularly in relation to motivation as a separate outcome, the effect of tempo, and realistic workplace activities. In doing so, this also points towards directions for future research.

1.4 Structure of the Thesis

The thesis proceeds as follows. Chapter 2 develops the theoretical framework. It presents the arousal–mood hypothesis, cognitive load theory, the limited-capacity tradition, the duplex-mechanism account of auditory distraction, and the cautionary history of the Mozart effect. These perspectives are then brought together into the activity-demand framework that organises the analysis. Chapter 3 sets out the methodology: the scoping review design, the eligibility framework, the search strategy across seven databases, and the procedures for selecting studies and charting data, reported in line with the PRISMA-ScR standard (Tricco et al., 2018). Chapter 4 presents the results, first as a numerical description of the included studies and then as a narrative organised by cognitive outcome, namely memory and recall, concentration and attention, reading and learning, and motivation. Chapter 5 discusses the main findings, their interpretation, their implications for educational and occupational practice, the gaps they reveal, and the strengths and limitations of the review. Chapter 6 draws the thesis together and returns directly to the research question.

Chapter 2: Background and Theoretical Framework

2.1 Overview of the Theoretical Framework

The conceptual and theoretical foundations of the thesis are established here. The aim is not to cover the full breadth of research on music and cognition, but to set out the main theoretical perspectives that help explain how, and under what conditions, music might be expected to help or to hinder cognitive performance. These perspectives are brought together because no single theory adequately explains the diverse findings reported in the empirical literature; each instead contributes to understanding a different mechanism through which music may influence cognitive performance. Interest in the psychological dimensions of music listening has a long history, with Mursell (1937) among the first to examine systematically how listeners respond to and are affected by music, and the field of music and emotion in particular has developed into a mature sub-discipline of psychology over the past two decades (Juslin & Sloboda, 2016). Taken together, these theoretical perspectives suggest that the effects of music are likely to vary according to the cognitive demands of the activity being performed. Music may support performance during relatively simple or routine activities but interfere with performance during cognitively demanding activities, although these patterns are also influenced by the characteristics of the music and by individual differences between listeners. Familiar but effortful activities may occupy a position between these two extremes. This activity-demand perspective is consistent with meta-analytic evidence that the effects of music vary across different task conditions (Kämpfe et al., 2011) and provides the conceptual framework used to organise and interpret the evidence presented in this review. These theories are used to explain music as a condition of study and work-related cognitive activity. The focus is therefore not listening to music in general, but the way music may shape learning, attention, motivation, and performance when another cognitive task is being performed.

The analysis proceeds as follows. It begins by examining the arousal–mood hypothesis and the inverted-U relationship between arousal and performance. Together, these perspectives help explain how music can facilitate performance, as well as the psychological mechanisms through which music influences emotional state. The account then turns to cognitive load theory and the limited-capacity model, which help to explain music's potentially disruptive effects. The Mozart effect is considered as a useful cautionary example of the risks of overstating music's cognitive benefits. The role of individual differences, including working-memory capacity, personality, music preference, and listening habits, is then discussed. The activity-demand framework is introduced as the organising structure for the analysis, distinguishing between routine activities, familiar but effortful activities, and truly novel or demanding activities across both study and work contexts. Finally, these perspectives are brought together to identify the gap in the literature that the present review seeks to address.

2.2 The Arousal–Mood Hypothesis and the Inverted-U

The arousal–mood hypothesis proposed by Thompson, Schellenberg, and Husain (2001) provides one theoretical explanation for why music may have beneficial effects on cognitive performance under certain conditions. According to this hypothesis, music does not enhance cognitive performance directly. Instead, it influences performance indirectly by altering the listener's level of arousal and mood, which in turn affect cognitive performance. Music that elevates mood and produces an appropriate level of arousal may enhance performance, whereas music that lowers mood or results in excessively high or low levels of arousal may instead impair performance. This hypothesis was developed partly to reinterpret the Mozart effect (see Section 2.5), relocating the source of any apparent benefit from the music itself to the affective state it induces. In their comprehensive review of the evidence, Schellenberg and Weiss (2013) concluded that the cognitive benefits of music listening are typically short-term and stem from music's impact on arousal and mood rather than from anything unique to music as such, noting that non-musical stimuli with a similar emotional impact can produce comparable effects.

The arousal–mood hypothesis is closely tied to the inverted-U relationship between arousal and performance, often associated with the Yerkes–Dodson tradition, which holds that performance is optimal at an intermediate level of arousal and declines when arousal is either too low or too high. This relationship is theoretically meaningful rather than merely descriptive: at low arousal the individual may be insufficiently alert or engaged, so attention drifts and engagement falls; at high arousal, attention narrows and processing capacity is consumed by the arousing stimulus itself. Performance is therefore best at the intermediate point at which the individual is sufficiently activated to sustain attention but not so activated that capacity is diverted. This optimal point is not fixed, but shifts with the difficulty of the activity, so that the more demanding the activity, the lower the level of arousal at which performance peaks. This inverted-U principle has a long history in the psychology of music, having been central to Berlyne's experimental aesthetics, in which listeners were held to prefer stimuli of intermediate arousal potential, determined by collative properties such as complexity, familiarity, and novelty (Hargreaves, 1986). The practical implication follows directly: the same piece of music may either support or disrupt performance depending on the listener's baseline arousal and on the demands of the activity. On a monotonous, under-stimulating activity, music can raise arousal toward the optimum and thereby improve focus; on an already demanding activity that places the listener near the top of the arousal curve, additional stimulation from music may push arousal past the optimum and degrade performance. Because the optimum sits lower for more demanding activities, there is less room for additional stimulation before performance begins to decline, which is why such activities are more susceptible to disruption. This single mechanism therefore predicts opposite effects of music for routine and for demanding activities, anticipating the central distinction of the review.

The arousal–mood hypothesis proposes that music influences cognitive performance through changes in emotional state, but it does not explain the psychological mechanisms through which music produces those changes. Contemporary music-emotion research addresses this question directly. Juslin and Sloboda (2016) propose that music induces emotion through a set of distinct underlying mechanisms, formalised in the BRECVEMA framework: brain-stem reflexes, rhythmic entrainment, evaluative conditioning, emotional contagion, visual imagery, episodic memory, musical expectancy, and aesthetic judgement. Several of these mechanisms are directly pertinent to the present review. Brain-stem reflexes help to explain why fast, loud, or rhythmically salient music raises physiological arousal more or less automatically; evaluative conditioning and episodic memory help to explain why preferred and familiar music produces stronger and more positive affective responses than unfamiliar or disliked music. The arousal and mood changes that the hypothesis treats as the proximate cause of cognitive effects are thus themselves the product of identifiable psychological processes, and the strength of those changes varies systematically with the characteristics of the music and the listener, a point developed further in the treatment of individual differences below (Juslin, 2016).

A particularly direct test of this mediation pathway is provided by Ba and Hu (2025), who examined how the tempo and mode of background music affected reading comprehension in graduate students, capturing emotion through self-report, facial-expression, and physiological measures. Tempo and mode did not influence comprehension directly, but the emotions the music induced did, so that the effect of the music on performance ran entirely through the listener's emotional response. This is close to a direct confirmation of the hypothesis's central claim, that music reaches cognition by way of emotional state rather than acting on it directly.

A related but distinct line of theorising, the Moderate Brain Arousal model (Sikström & Söderlund, 2007), proposes that moderate auditory stimulation benefits cognitive performance through a stochastic resonance mechanism modulated by dopamine transmission, with the model's central prediction concerning individuals with attention-deficit/hyperactivity disorder, who are theorised to require greater auditory stimulation than neurotypical individuals to reach an optimal performance state. Because this model was developed to explain noise effects in clinical populations rather than music's effects in healthy listeners, it falls outside the Population and Concept criteria defined for this review and is not pursued further here.

2.3 Cognitive Load Theory and the Limited-Capacity Model

Where the arousal–mood hypothesis explains how music can help, cognitive load theory and the limited-capacity model explain how it can hinder. Cognitive load theory (Sweller, 1988) holds that working memory is a limited resource and that performance on a learning or problem-solving activity suffers when the total cognitive load exceeds that capacity. From this perspective, background music may impose an additional, extraneous load: processing the music consumes finite resources also needed for the primary activity, leaving less available for the activity itself. This concern follows from the concept of background music itself, since, as Schellenberg and Weiss (2013) observe, the term implies that listeners are attempting to do two things at once, so that the cognitive limitations of a dual-task situation are likely to play a role and may lead to reductions in performance on the primary activity.

This argument has a long history in the limited-capacity tradition initiated by Broadbent (1958), whose filter model held that the human information-processing system can attend to only a limited amount of incoming information at once, and must therefore select some inputs for further processing while filtering others out. On this view, attention functions as a limited-capacity filter: when the quantity of incoming information exceeds the system's capacity, some of it is necessarily lost. Background music adds to the incoming stream, and when music and a cognitive activity compete for the same limited channel, one or both must suffer. The degree of competition, however, depends on how far the music and the activity draw on the same kind of processing rather than simply on how much information each carries. This qualification is what moves the explanation beyond an overly simple quantity-of-noise model: it predicts that interference is selective, falling most heavily where music and activity make demands on the same specialised resources. This distinction is central here, because it predicts that music will be most disruptive when it engages the very processes the activity demands. A reading-comprehension activity, which depends on verbal and semantic processing, should therefore be especially vulnerable to music that contains verbal, semantic content (that is, to lyrics), whereas instrumental music, lacking semantic content, should interfere far less even at the same volume. The prediction is thus not that music is distracting in general, but that music with verbal or semantic content is especially likely to interfere with activities that also require verbal or semantic processing.

This prediction is given its most precise contemporary formulation in the duplex-mechanism account of auditory distraction (Hughes, 2014). Hughes distinguishes two functionally distinct forms of auditory distraction. *Interference-by-process* arises when the involuntary processing of a sound competes with a similar process that the listener is deliberately applying to a focal activity. The key feature is that the competing process is the same in kind: the sound is not merely present but recruits the very operation the activity depends upon, so that the two draw on a shared, specialised resource rather than on general attention. On this account, semantically meaningful lyrics interfere with reading precisely because both require semantic processing, and the interference is obligatory, because the processing of intelligible speech cannot easily be switched off at will. Attentional capture, by contrast, arises when a sound disengages attention from the activity regardless of the processes involved, as when a salient or personally significant sound (one's own name, or one's native language) draws attention away. Importantly, Hughes argues that attentional capture can be controlled through greater top-down task-engagement, whereas interference-by-process cannot. This framework helps to explain both why lyrical music is so reliably disruptive to verbal activities and why some listeners, through habit and engagement, are better able to resist distraction. Meta-analytic evidence is consistent with this picture: in a Bayesian meta-analysis of 65 studies, Vasilev et al. (2018) found that background noise, speech, and music each produced a small but reliable impairment of reading performance, with the disruption attributable in large part to the semantic and linguistic content of the competing sound rather than to its mere presence.

A related account, the seductive detail effect, extends the same reasoning to learning from text. Seductive details are interesting but extraneous elements that draw attention away from the material to be learned; on this view, background music functions as an auditory seductive detail that competes for limited working-memory resources. Lehmann and Seufert (2017) tested this alongside the arousal–mood hypothesis and found no evidence that arousal or mood mediated music's effect on learning but did find that working-memory capacity interacted with background music for comprehension, such that learners with higher working-memory capacity comprehended better with music than without. This pattern is consistent with the seductive detail account and indicates that the capacity to accommodate an additional, extraneous load may itself determine whether music helps or hinders learning.

2.4 Attention, Concentration, and Controlled Processing

A brief clarification of the term concentration is warranted before the theoretical perspectives are drawn together, since concentration is one of the cognitive outcomes at issue here. In this work, concentration is understood not as a single, unitary ability but as a combination of attentional processes that allow the listener to remain engaged with the activity, filter out irrelevant stimulation, and maintain task-relevant information in working memory (Posner & Petersen, 1990). Background music may affect these processes in different ways and to different degrees. For low-demand or repetitive activities, music may help to sustain alertness and reduce mind-wandering by raising arousal to a more suitable level. For demanding activities, however, concentration depends more heavily on controlled attention and working memory, leaving less capacity available for processing irrelevant sound.

Classic accounts of attention and working memory make the same prediction. Kahneman (1973) characterised attention as a limited resource that must be allocated across competing demands according to the difficulty of the activity and the listener's level of arousal, a framing that anticipates precisely the demand-dependent effect described above. The vulnerability of verbal activity in particular is well explained by the structure of working memory. Baddeley and Hitch (1974) proposed that verbal material is maintained by a dedicated phonological component, which means that spoken or sung words compete directly with the phonological processes involved in reading, rehearsal, and verbal learning. Lyrics, on this account, are not disruptive merely because they add sound, but because they draw on the very verbal resources that the primary activity also requires, a conclusion that converges with the interference-by-process mechanism described above (Baddeley & Hitch, 1974; Hughes, 2014). Finally, the contrast between automatic and controlled processing helps to explain why routine activities are less vulnerable to music than novel or complex ones. Schneider and Shiffrin (1977) showed that well-practised activities come to rely on automatic processing, which places few demands on controlled attention, whereas unfamiliar activities require more deliberate monitoring and are therefore more easily disrupted by competing auditory input.

2.5 The Mozart Effect: A Cautionary Example

The Mozart effect provides an important reference point in the study of music and cognition: it is the claim that brief exposure to music by Mozart produces a short-term improvement in spatial-reasoning performance. The effect entered both the scientific literature and popular culture following Rauscher, Shaw, and Ky (1993), who reported that participants performed better on spatial tasks after listening to a Mozart sonata than after silence or relaxation instructions. The finding was rapidly and enthusiastically generalised in the popular media into the broad claim that listening to Mozart makes people, and especially children, more intelligent.

Subsequent research has provided limited support for this claim. The effect proved notoriously difficult to replicate, and the most comprehensive assessment to date, a meta-analysis by Pietschnig, Voracek, and Formann (2010) synthesising approximately 40 studies and more than 3,000 participants, found little support for any specific cognitive benefit attributable to Mozart's music as such. As Schellenberg and Weiss (2013) document in their review, the most rigorous tests of the phenomenon found that when differences in arousal and mood between conditions were held constant by statistical means, the apparent Mozart advantage disappeared entirely; moreover, enjoyable music of other kinds, and even non-musical stimuli of comparable emotional impact, produced equivalent benefits. Where small effects appeared, then, they were consistent with the more general arousal–mood hypothesis rather than with anything unique to Mozart. The Mozart effect is included here not as evidence of music's cognitive benefits but as a cautionary example: it illustrates how readily a modest, indirect, arousal-mediated effect can be over-interpreted as a direct and durable enhancement of intelligence. This caution informs the conservative, mechanism-focused stance adopted throughout this thesis.

Neurophysiological work has offered a parallel line of evidence on the priming interpretation of the effect. Jaušovec et al. (2006), recording EEG during spatial-rotation problem-solving after exposure to Mozart's Sonata K.448, reported improved task performance and less complex cortical activity following the music, which they interpreted as support for a priming explanation. Because this paradigm involves listening before rather than during the task, it lies outside the scope of the present review's focus on background music accompanying an activity, but it illustrates the enduring scientific interest in the mechanisms underlying the effect.

2.6 Individual Differences

A consistent finding of the music-and-cognition literature is that the effect of background music is not uniform across listeners. Three sources of individual difference are particularly relevant and are repeatedly reflected in the evidence charted in the results chapter.

2.6.1 Working-Memory Capacity

If music disrupts performance by consuming limited working-memory resources, then individuals with greater working-memory capacity should be better able to absorb the additional load without cost. This prediction follows directly from cognitive load theory and the limited-capacity model, and it is supported in the empirical literature: individuals with lower working-memory capacity tend to be more vulnerable to disruption by background music on demanding activities, while those with higher capacity are less affected (Christopher & Shelton, 2017). Working-memory capacity thus functions as a moderator of the activity-demand pattern, influencing how strongly music disrupts performance on high-load activities.

2.6.2 Personality: Introversion and Extraversion

Eysenck's theory of personality links introversion and extraversion to differences in baseline cortical arousal, with introverts characteristically more highly aroused than extraverts (as discussed in Dobbs et al., 2011). On the logic of the inverted-U, the same background music should therefore push introverts and extraverts to different points on the arousal curve: extraverts, with lower baseline arousal, may benefit from the additional stimulation that music provides, whereas introverts, already closer to their optimum, may be tipped past it into impairment. This prediction has substantial empirical support; introverts have repeatedly been found to be more disrupted by background music and noise than extraverts on demanding cognitive activities (e.g., Dobbs et al., 2011), and several of the charted studies likewise report stronger benefits of music among extraverts.The prediction is therefore two-sided: the same piece of music may move an introvert past the optimum into impairment while moving an extravert, who begins from lower baseline arousal, toward the optimum, aiding performance or leaving it unharmed. Personality does not simply make some listeners more vulnerable to music; it helps determine the direction in which a given piece of music shifts a listener along the arousal curve.

2.6.3 Music Preference and Listening Habit

Finally, whether a listener prefers the music being played, and whether they are habituated to studying with music, shapes its effect. The development and determinants of musical preference have been studied extensively, and any account of preference must consider the interaction of the listener, the music, and the listening situation (Greasley & Lamont, 2016; Hargreaves, 1986). Preferred music lifts mood and arousal more strongly than unfamiliar or disliked music, as the arousal–mood hypothesis predicts. Familiarity is itself a contributing factor: repeated exposure renders music more predictable and therefore less attentionally demanding, so that familiar music competes less for the resources a focal activity requires (Margulis, 2014); this is consistent with the broader principle that mere exposure tends to increase liking for a stimulus. Habitual listeners, moreover, appear better able to resist distraction, consistent with Hughes's (2014) claim that top-down task-engagement can attenuate attentional capture. Listeners also use music purposefully, selecting it to regulate their arousal, mood, and motivation as they work (Sloboda et al., 2016). The charted evidence indicates that listeners accustomed to studying with music are less impaired by it than non-listeners, and preference can determine whether music helps or harms on otherwise identical activities. Preference and habit are therefore not secondary or incidental factors but central to understanding the heterogeneity of findings in this field.

2.7 The Activity-Demand Framework: From Routine to Novel Activities

These theoretical perspectives can now be integrated into the activity-demand framework that structures the analysis. The arousal–mood hypothesis and the inverted-U explain music's capacity to *help*; cognitive load theory, the limited-capacity model, and the duplex-mechanism account explain its capacity to *hinder*. A key factor that determines which tendency dominates is the cognitive demand of the activity being undertaken. Rather than a simple opposition between routine and demanding work, it is helpful to think of activities as occupying a spectrum of demand, on which three broad positions can be distinguished. This distinction applies equally to studying and to work, since the cognitive demand of an activity, rather than the setting in which it occurs, is what determines how music is likely to affect it. In both study and work settings, some activities are new and require the acquisition of knowledge or skill, while others are familiar in the sense that the necessary knowledge or skill is already possessed, yet still call for active and effortful performance.

The distinction between routine and novel activities can be understood more precisely through the difference between automatic and controlled processing. Schneider and Shiffrin (1977) showed that well-practised activities come to rely on automatic processing, which proceeds with little demand on attentional control and without straining the capacity limits of the system, whereas novel or variable activities require controlled processing, which is effortful, capacity-limited, and dependent on active monitoring. This difference helps to explain why the position of an activity on the demand spectrum is not arbitrary. Because routine activities draw mainly on automatic processing, they leave controlled capacity free, so that background music may be tolerated or may even support performance by maintaining arousal and motivation. Novel and complex activities, by contrast, depend on controlled processing and therefore compete with music for the same limited resources, which is why music is expected to become more disruptive as activities move toward the demanding end of the spectrum.

At one end of the spectrum lie routine activities: repetitive, well-practised, and largely automatic work such as sustained attention, vigilance, data entry, or the rehearsal of already-mastered material. These activities place little demand on working memory. The listener is typically under-stimulated, and the main risk to performance is mind-wandering and disengagement. Here music's arousal-raising and mood-lifting properties are an asset: by moving the listener up the arousal curve toward the optimum, music can sustain attention and reduce mind-wandering without competing for scarce task-specific resources, because such activities make few demands on those resources. Research on the everyday uses of music indicates that listeners often recruit music for this purpose, using it to maintain arousal and motivation during otherwise tedious activities (Sloboda et al., 2016).

In the middle of the spectrum lie familiar but effortful activities: those for which the listener already possesses the relevant knowledge or skill, but which must still be actively performed and therefore continue to draw on working memory. Completing a set of practice problems of a familiar type, applying a known procedure, or writing up material that has already been understood would fall into this category. Such activities are new in the sense that they must be done afresh on each occasion, yet they are not novel in the sense of requiring new learning. Their intermediate demand makes the effect of music correspondingly less predictable: where the activity leaves some capacity free, music may be tolerated or even helpful, but where it approaches the limits of available resources, music begins to compete for them. The position of a given activity within this middle band, and the characteristics of the music and the listener, together determine whether music helps, hinders, or has little effect.

At the far end of the spectrum lie novel, high-demand activities: genuinely new and knowledge-demanding work such as reading comprehension, learning unfamiliar material, verbal memorisation, and complex reasoning. In these activities, listeners are already operating near the limits of working memory, so the main challenge is competition for cognitive resources. Music's demand on shared processing resources becomes a liability, particularly when the music contains semantically rich content (lyrics) that competes directly with the verbal processing the activity requires. It is on activities of this kind that music, and especially lyrical music, is most likely to impair performance.

This graded framework yields the central expectation that the thesis then examines against the evidence: that the effect of background music shifts systematically along the demand spectrum, tending to aid or to leave unharmed performance on routine activities, to produce mixed and conditional effects on familiar but effortful ones, and to impair performance on novel, high-demand activities. Kämpfe et al. (2011), in a meta-analysis of the effects of background music on adult listeners, provided explicit support for this broad task-dependent pattern, lending meta-analytic weight to a distinction that might otherwise rest only on theoretical inference. At the same time, the framework is advanced here as a heuristic rather than a law. The individual-difference moderators discussed above ensure that the pattern is probabilistic, not deterministic, and, as the evidence reviewed later indicates, there are instructive exceptions in which instrumental music aids even demanding activities. The value of the framework lies not in predicting every case but in providing a principled basis for mapping, interpreting, and making sense of an otherwise highly inconsistent body of literature, across both study and work.

2.8 Theoretical Synthesis and Research Gap

Together, these theoretical perspectives help to explain why findings on music and cognition have often appeared inconsistent. Music exerts no single, uniform effect on cognitive performance. Rather, it acts through two opposing tendencies, an arousal- and mood-mediated facilitation on the one hand and a resource-competition-driven interference on the other, with the overall effect depending on the demands of the activity, the content of the music, and the characteristics of the listener. The apparent contradictions in the literature, in which music is reported to help in some studies and to harm in others, are thus largely reconciled once the moderating role of activity type, lyrical content, and individual differences is considered.

What remains less well integrated is the literature itself. Although the individual theoretical mechanisms are well established, and although meta-analytic evidence supports the broad task-dependent pattern, the empirical studies are dispersed across diverse populations, settings, musical conditions, and cognitive outcomes, and they have rarely been mapped together in a way that makes the overall shape of the evidence visible. The breadth and heterogeneity of this literature, which spans memory, learning, motivation, and concentration, and which ranges across study and work activities from laboratory experiments to naturalistic classroom and workplace settings, make it well suited to a scoping review, whose purpose is precisely to map a wide and varied body of evidence, to clarify key concepts, and to identify gaps rather than to synthesise toward a single effect estimate. It is this mapping that the present review undertakes, organising the evidence through the activity-demand framework set out above and attending throughout to the moderating roles of musical characteristics and individual differences.

Chapter 3: Methodology

3.1 Research Design

This study used a scoping review methodology to map existing empirical research on the effects of music on memory, learning, motivation, and concentration during study and work-related cognitive tasks. A scoping review was selected rather than a systematic review because the research question is broad and exploratory, encompassing multiple cognitive outcomes, diverse populations, and a range of musical characteristics. Whereas systematic reviews are usually suited to more narrowly defined questions and to synthesising evidence in relation to a specific effect or question, scoping reviews are designed to map the breadth of heterogeneous literature, clarify key concepts, and identify gaps in the evidence base (Arksey & O'Malley, 2005; Munn et al., 2018). Given that the literature on music and cognition is characterised by methodological diversity and inconsistent findings, this approach was considered appropriate for mapping the field in a broad and transparent way.

The review was guided by the methodological framework originally proposed by Arksey and O'Malley (2005) and subsequently refined by Levac et al. (2010) and was reported in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses extension for Scoping Reviews (PRISMA-ScR; Tricco et al., 2018). The framework comprises five stages: (a) identifying the research question; (b) identifying relevant studies; (c) selecting studies; (d) charting the data; and (e) collating, summarising, and reporting the results. Each stage is described in turn below.

3.2 Research Question and the PCC Framework

The review was guided by the following research question: How does music affect memory, learning, motivation, and concentration during study and work-related cognitive tasks, and under what conditions do these effects occur? In keeping with established guidance for scoping reviews, the eligibility of studies was framed using the Population, Concept, and Context (PCC) framework recommended by the Joanna Briggs Institute (Peters et al., 2020). The application of the PCC framework to the present review is summarised in Table 3.1.

Table 3.1

Application of the PCC Framework to the Review Eligibility Criteria

Element	Definition applied in this review
Population	Non-clinical children, adolescents, and adults, including students and general healthy learners. Clinical populations (e.g., individuals with dementia, clinical ADHD samples receiving treatment, or patients undergoing medical procedures) were excluded.
Concept	Music as an intervention, condition, or exposure — including background music, preferred and non-preferred music, instrumental versus vocal music, and variations in tempo, genre, and lyrical content — and its effect on memory, recall, learning, motivation, concentration, and cognitive task performance.
Context	Learning, studying, and cognitive task-performance settings, including laboratory, classroom, and naturalistic study environments.

3.3 Eligibility Criteria

Inclusion and exclusion criteria were defined a priori to ensure that study selection was transparent and reproducible. Studies were included if they satisfied all of the following criteria:

- They were empirical studies that were charted as primary evidence; relevant systematic reviews and meta-analyses were retained as higher-level contextual evidence rather than charted (see Section 3.6);
- They examined music as an intervention, condition, or independent variable;
- They involved a non-clinical human population of any age;
- They measured at least one outcome relevant here, namely memory or recall, learning, academic performance, motivation, concentration, or cognitive task performance;
- They were published in English; and
- They were published in or after the year 2000, with seminal earlier works retained for contextual and theoretical discussion only.
- Studies were excluded if they met any of the following criteria:
- They focused exclusively on clinical populations or on music therapy as a clinical intervention;
- They examined music exclusively in non-cognitive performance contexts, such as driving, physical exercise, or consumer behaviour;
- They concerned music's role in emotional self-regulation, autobiographical memory, or identity formation rather than its effect on cognitive or academic performance;

- They investigated the effects of musical training or instrumental instruction on cognitive or neural development, as this constitutes a distinct body of literature;
- They were opinion pieces, editorials, conference abstracts, or other non-peer-reviewed sources; or
- They were published in a language other than English.

Peer-reviewed empirical studies were included in the charted evidence. Relevant systematic reviews and meta-analyses were retained as higher-level contextual evidence and were used to inform the theoretical and interpretive discussion, but they were not included in the primary-study count. This distinction allowed the review to map the empirical evidence directly while still situating the findings within the wider review-level literature.

3.4 Information Sources and Search Strategy

A structured search was conducted across seven electronic databases selected for their comprehensive coverage of the education, psychology, and cognitive neuroscience literatures: PubMed/MEDLINE, PsycINFO, the Education Resources Information Center (ERIC), Scopus, Web of Science, Google Scholar, and the Cochrane Library. The search was conducted to identify eligible studies comprehensively and to confirm those already located during the earlier stages of the review. Because several databases returned very large result sets, results from these were sorted by relevance and a fixed number of the highest-ranked records was screened, in line with common practice for managing high-yield sources: the first 100 records were screened for PsycINFO and Scopus, the first 50 for Google Scholar, which alone returned tens of thousands of records of widely varying relevance, and the first 30 for ERIC. Smaller result sets were screened in full. In addition, a manual search was performed of the journal *Educational Research Review* (Elsevier) to ensure comprehensive coverage of review-level evidence within the field of education, and the reference lists of included studies were screened to identify further relevant sources (a technique known as backward citation searching). Screening of Google Scholar surfaced no eligible studies beyond those already identified through the other databases and citation searching, confirming that the core studies were also the most relevant records returned by a broad search.

The searches were conducted between 20 February and 7 July 2026, with the later searches run to update coverage and capture the most recent eligible evidence. Search terms were developed iteratively and combined using the Boolean operators AND and OR, together with truncation and database-specific controlled vocabulary where available. The search strategy combined three conceptual blocks corresponding to the PCC elements: terms relating to music, terms relating to cognitive and academic outcomes, and terms relating to the population. The primary search string is reproduced below, and the full search strings as adapted for each database are provided in Appendix A.

(“background music” OR “music listening” OR “instrumental music” OR “music exposure”) AND (“concentration” OR “attention” OR “cognitive performance” OR “task performance” OR “memory” OR “recall” OR “learning” OR “motivation” OR “academic performance”) AND (“students” OR “pupils” OR “learners” OR “children” OR “adolescents” OR “university students” OR “workers” OR “employees” OR “workplace” OR “occupational”)

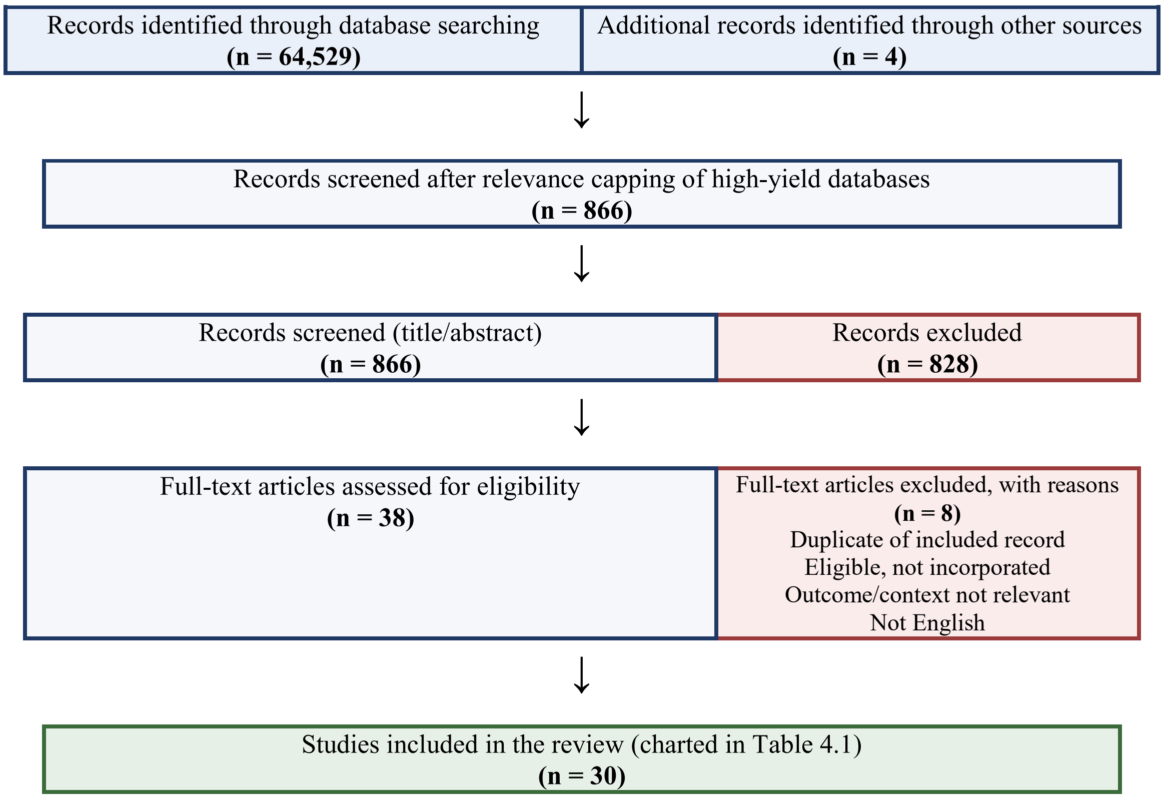
Because the educational literature predominates in this field, several of the workplace and work-analogue studies were identified through backward citation searching and the manual journal search rather than through the database search alone. This is noted here because it bears on the comparatively small number of occupational studies in the charted set, a point returned to in the discussion of limitations.

3.5 Study Selection

All records retrieved from the database searches were exported to reference management software, where duplicate records were identified and removed. The study selection process proceeded in two stages. In the first stage, the titles and abstracts of all unique records were screened against the eligibility criteria. In the second stage, the full texts of all potentially relevant records were retrieved and assessed in detail. Records excluded at the full-text stage were documented together with the reason for their exclusion. The selection process is reported in full below in the PRISMA-ScR flow diagram (Figure 3.1), which records the number of records identified, screened, assessed for eligibility, and ultimately included. A total of 30 studies met the eligibility criteria and were included in the review. As it was conducted by a single researcher, screening and charting were carried out by one reviewer rather than by two independent reviewers; the implications of this for the reliability of selection are considered later among the limitations.

Figure 3.1

PRISMA-ScR Flow Diagram of the Study Identification, Screening, and Selection Process



Note. Database searches returned 64,529 records in total (PubMed = 541; PsycINFO = 1,605; Scopus = 17,273; Web of Science = 12; ERIC = 769; Cochrane Library = 29; Google Scholar ≈ 44,300). Because PsycINFO, Scopus, and Google Scholar returned very large result sets, these were sorted by relevance and screened to a fixed depth (the first 100 for PsycINFO and Scopus; the first 50 for Google Scholar; the first 30 for ERIC); smaller sets were screened in full. The "records screened" figure (n = 866) therefore reflects the records actually examined rather than the full number returned. The 866 screened records comprise 862 retained from database searching after relevance capping and the 4 records identified through other sources. The included count (n = 30) corresponds to the studies charted in Table 4.1

3.6 Data Charting

In accordance with scoping review methodology, data were charted rather than statistically synthesised. A data-charting form was developed in a spreadsheet to record the following information from each included study: author(s) and year of publication; country in which the study was conducted; the population and sample characteristics; the music conditions examined; the cognitive or academic outcomes measured; the principal findings; and any reported moderating variables, such as personality or individual differences. The charting form was developed iteratively: it was piloted on a small number of included studies and refined to ensure that it captured the information necessary to address the research question. The charted data are presented in tabular form and provide the basis for the narrative summary of findings.

In addition to the descriptive information above, each study was coded according to the activity-demand framework. The primary cognitive task in each study was classified as routine, familiar but effortful, or novel and high-demand, with a fourth category, mixed, used where a study employed a battery of tasks or self-report measures that spanned more than one level. Classification was based on the cognitive demand of the primary task rather than on the setting in which it was performed, in keeping with the central argument that the demand of the activity, not its context, governs the effect of music. Borderline cases were classified conservatively and assigned to the mixed category where a study included several task types of differing demand; such cases are identified in the notes to Table 4.1 so that the basis of the classification remains transparent.

3.7 Collating, Summarising, and Reporting the Results

Consistent with the final stage of the Arksey and O'Malley (2005) framework, the charted data were collated and summarised both numerically and narratively. The numerical summary describes the distribution of studies according to characteristics such as year of publication, population, type of music examined, and outcomes measured. The narrative summary organises the findings thematically according to the principal cognitive outcomes addressed in the research question, namely memory and recall, learning and comprehension, concentration and attention, and motivation, and considers the musical characteristics (such as tempo, lyrical content, and listener preference) associated with supportive or disruptive effects. This thematic organisation is intended to map the evidence in a manner that directly addresses the research question and that highlights both points of consensus and areas of inconsistency within the literature.

3.8 A Note on Quality Appraisal

In contrast to systematic reviews, scoping reviews do not usually require a formal appraisal of the methodological quality of included studies, as the objective is to map the breadth of the available evidence rather than to weigh the strength of that evidence toward a definitive conclusion (Arksey & O'Malley, 2005; Tricco et al., 2018). Accordingly, no formal quality appraisal was undertaken in the present review. Nevertheless, methodological characteristics relevant to the interpretation of findings, such as sample size, study design, and the nature of the comparison condition, were recorded during data charting and are considered in the discussion of the findings, so that the mapped evidence can be interpreted on an informed basis.

3.9 Ethical Considerations

As this study constituted a review of previously published research and did not involve the collection of data from human participants, formal ethical approval was not required. The review was nonetheless conducted in accordance with the principles of research integrity, including the transparent and accurate reporting of methods and findings and the appropriate acknowledgement of all sources.

Chapter 4: Results

4.1 Overview of the Charting Process

Consistent with the methodology already described, the included studies were charted rather than formally synthesised, and the charted data are reported both numerically and narratively (Arksey & O'Malley, 2005; Levac et al., 2010). The numerical description, presented first, describes the distribution of the included studies according to their principal characteristics. The narrative summary that follows is organised around the four cognitive outcomes specified in the research question, namely memory and recall, concentration and attention, reading and learning, and motivation, and within each outcome the evidence is examined through the activity-demand framework. The full charted data for all included studies are presented in Table 4.1, and the selection process by which they were identified is reported in the PRISMA-ScR flow diagram (Figure 3.1).

The findings are presented using the activity-demand framework developed in Section 2.7, which distinguishes routine, familiar but effortful, and novel, high-load activities, and which received meta-analytic support from Kämpfe et al. (2011). It is applied here as an organising guide rather than a fixed classification, because several included studies employed sets of tasks that span more than one level. The narrative notes both where the pattern is supported and where the evidence is more mixed, in keeping with the mapping purpose of a scoping review.

Table 4.1 records, for each included study, the authors and year, the population and sample, the music conditions examined, the activity-demand level, the outcomes measured, and the principal findings. The patterns visible in the table are described numerically in the next section and analysed thematically in the sections that follow.

Table 4.1Characteristics and Principal Findings of Included Studies, Classified by Activity-Demand Level

#	Author(s) & Year	Population / Sample	Music Condition(s)	Task Type (activity-demand level)	Outcome(s) Measured	Key Finding
1	*Anderson & Fuller (2010)*	334 US 7th–8th grade students	Lyrical pop vs. silence	Novel / high-load (reading comprehension)	Reading comprehension	Comprehension declined significantly with lyrical music; stronger decline for those preferring to study with music.
2	*Dobbs, Furnham & McClelland (2011)*	118 female secondary students (UK)	Garage music vs. office noise vs. silence	Novel / high-load (reasoning, ability tests)	Abstract reasoning, general ability, verbal reasoning	Best in silence; introverts more impaired by music and noise than extraverts.
3	*Avila, Furnham & McClelland (2012)*	58 students aged 16–18 (UK)	Vocal vs. instrumental vs. silence	Novel / high-load (verbal, numerical, logic)	Verbal, numerical, logic tests	Verbal performance best in silence; lyrics interfered with verbal processing.
4	*Doyle & Furnham (2012)*	54 adults (27 creative)	Music (120–130 bpm) vs. silence	Novel / high-load (reading comprehension)	Reading comprehension	No significant interaction; trend for creative individuals to be less disrupted by music.
5	*Johansson et al. (2012)*	Adult participants (Sweden)	Preferred vs. non-preferred study music	Familiar but effortful (reading with existing skill)	Reading comprehension + eye movements	Music preference influenced reading processing measured via eye-tracking.
6	*Ferreri et al. (2013)*	Healthy older adults	Background music vs. silence	Novel / high-load (verbal encoding)	Verbal memory; prefrontal activity (fNIRS)	Music reduced prefrontal effort while improving verbal memory ('less effort, better results').
7	*Christopher & Shelton (2017)*	Students (healthy adults)	Background music vs. control	Familiar but effortful (working-memory task)	Cognitive performance × working-memory capacity	Lower working-memory individuals more negatively affected by background music.
8	*Nguyen & Grahn (2017)*	Healthy adults (3 experiments)	Music varied in mood & arousal vs. silence	Familiar but effortful (memory tasks)	Recall, recognition, associative memory	Low-arousal music improved recall and recognition vs. high-arousal music.

#	Author(s) & Year	Population / Sample	Music Condition(s)	Task Type (activity-demand level)	Outcome(s) Measured	Key Finding
9	*Kiss & Linnell (2021)*	40 students	Preferred music vs. silence	Routine (low-demand vigilance task)	Sustained attention; attentional state	Preferred music increased task focus and reduced mind-wandering on a low-demand task.
10	*Kiss & Linnell (2024)*	Adults habituated to study music (2 experiments)	Preferred music vs. silence; vs. noise	Routine (low-demand vigilance task)	Sustained attention; mood; arousal	Music decreased mind-wandering, increased task focus, speeded RT; mood and arousal mediated effects.
11	*Souza & Barbosa (2023)*	College students (Psychology), Univ. of Porto, aged 18–58; within-subjects (Ns: verbal 113, visual 118, reading 123, arithmetic 106)	Silence vs. instrumental (lo-fi hip-hop) vs. music with lyrics (pop)	Novel / high-load (memory, reading, arithmetic)	Verbal/visual memory, reading, arithmetic	Lyrics hindered verbal memory, visual memory and reading ($d \approx -0.3$); arithmetic effect not credible; instrumental music neutral. Participants recognised the lyrics' cost but tended to misperceive instrumental music as beneficial.
12	*Moreno & Woodruff (2024)*	74 first-year undergraduates, aged 18–21 (termed 'adolescent learners' by the authors), Canada	Fast (150 bpm) vs. slow (110 bpm) vs. no music; Mozart K.448, tempo-manipulated instrumental	Novel / high-load (reading comprehension)	Reading comprehension (Nelson-Denny); facial-emotion expression; skin-conductance	Fast tempo lowered comprehension ($M = 3.61$) and raised fear/contempt and skin-conductance; slow tempo produced highest scores ($M = 4.04$); no-music intermediate ($M = 3.85$).
13	*Lin, Kuo & Mai (2023)*	51 analysed (52 recruited); Vietnamese undergraduates, mean age 23.7	Fast (138 bpm) vs. slow (66 bpm) vs. no music; Mozart K.448 Andante, tempo-manipulated	Routine (processing-speed tasks)	Cognitive processing speed (motor, visuospatial, linguistic)	Slow tempo worsened processing speed across all three tasks vs. no music; fast tempo did not differ from no music; slow tempo improved accuracy on the hardest (linguistic) task. Arousal did not mediate the effect.

#	Author(s) & Year	Population / Sample	Music Condition(s)	Task Type (activity-demand level)	Outcome(s) Measured	Key Finding
14	*Vigl et al. (2023)*	Students (classroom; mixed methods)	Self-selected classroom music	Mixed (classroom learning activities)	Mood, motivation, concentration, learning	Positive high-energy music improved mood, motivation and learning; effects varied by individual.
15	*Sun et al. (2024)*	90 Chinese college students (50 female), aged 18–21; listeners (n = 45) / non-listeners (n = 45)	Pop music with lyrics: Mandarin (L1) vs. English (L2) vs. no music	Novel / high-load (reading comprehension)	Reading comprehension (L1 & L2)	Music with lyrics reduced comprehension vs. no music; same-language lyrics most disruptive; non-listeners more impaired than listeners; native-language (Mandarin) lyrics worst overall. Supports duplex-mechanism account.
16	*Su, He & Li (2023)*	60 analysed (of 77) Chinese EFL sophomores, English majors (China)	Instrumental (Bach), fast (120 bpm) vs. slow (60 bpm); music preference high vs. low	Familiar but effortful (reading in a known second language)	English reading comprehension + eye movements	Fast-tempo music produced faster reading than slow tempo; tempo effect greater on easy texts; high-preference listeners outperformed low-preference; low-preference readers most hindered on difficult texts with slow tempo.
17	*Shih, Huang & Chiang (2012)*	Adults (work-like setting)	Background music conditions vs. control	Routine (work-like attention task)	Attention performance	Background music conditions related to attention performance; depends on music type and exposure.
18	*Huang & Shih (2011)*	Workers (healthy adults)	Background music vs. none; music type	Routine (repetitive work tasks)	Concentration / work performance	Music affected worker concentration; preference and type influenced whether helpful or harmful.

#	Author(s) & Year	Population / Sample	Music Condition(s)	Task Type (activity-demand level)	Outcome(s) Measured	Key Finding
19	*Taheri et al. (2022)*	52 university students (26 M / 26 F), aged 18–30 (Iran)	Classical instrumental (Clayderman) vs. silence	Mixed (2-back familiar-but-effortful; CPT, pegboard, coordination routine)	Working memory (2-back); sustained attention (CPT); manual dexterity; bimanual coordination	Music improved working memory and skill/dexterity performance but had no significant effect on sustained attention; effects stronger in extraverts.
20	*Goltz & Sadakata (2021)*	General participants (survey)	Self-reported study-music habits	Mixed (self-reported study activities)	Perceived focus and cognitive performance	Maps how and why people use music while studying; real-world context rather than experimental effect.
21	*Cooke, Speelman & Hollett (2026)*	226 university students, mean age 28.41 (range 18–55), 83% female (Australia); 157–161 for working-memory analyses	Self-reported listening vs. avoiding; genre, tempo and lyrical preferences while reading	Mixed (self-reported reading/study habits)	Music-listening habits; perceived helpfulness; working memory (OSpan); trait mind-wandering; music engagement	54% listen while reading for study, 46% avoid; listeners preferred non-lyrical, slow-tempo music. Working-memory capacity and mind-wandering were not associated with listening; music engagement was associated with listening and perceived helpfulness.
22	*Scott, Awasty, Li, Conlon, Johnson, Voorhees & Passantino (2025)*	108 employees (ESM pilot); 252 undergraduates (lab); 247 employees (3-week ESM), USA	Amount of time spent listening to music at work; within-person manipulation (little/no vs. 1 hr vs. 3 hr)	Mixed (workplace proofreading and self-rated work tasks)	Task attentional focus; objective and self-rated task performance; coworker ratings; willpower belief	Time spent listening to music related to performance in an inverted-U pattern rather than linearly; the curvilinear cost of extended listening was more pronounced among those believing willpower to be limited. Robust multi-study workplace evidence.

#	Author(s) & Year	Population / Sample	Music Condition(s)	Task Type (activity-demand level)	Outcome(s) Measured	Key Finding
23	*Que, Zheng, Hsiao & Hu (2023)*	100 non-native English speakers (between-subjects)	Self-selected background music vs. silence during reading	Novel / high-load (reading comprehension)	Reading comprehension; metacognition; eye movements; WMC, proficiency, personality	No difference in comprehension accuracy; eye-tracking showed music increased cognitive load at post-lexical (not lexical) stages. Higher proficiency and habitual use reduced load; higher WMC was associated with greater invested effort.
24	*Lehmann & Seufert (2017)*	81 college students (between-subjects, Germany)	Two pop songs (lyrical) vs. silence while learning a visual text	Novel / high-load (learning and comprehension)	Recall; comprehension; arousal and mood (mediators); working-memory capacity	No mediation by arousal or mood; no main effect on recall; interaction on comprehension such that higher working-memory capacity was associated with better learning with music. Consistent with the seductive-detail account.
25	*Sun, Y. (2025)*	428 Chinese undergraduates (two-phase RCT)	Control vs. high-arousal vs. low-arousal vs. Mozart K.448 (structured instrumental)	Familiar but effortful (Backward Digit Span working memory)	Task performance via flow and work engagement (mediators); 30-day habituation	Structured instrumental music supported performance indirectly through flow and engagement; high-arousal music was detrimental. The benefit attenuated by ~53% after 30 days of familiarity but did not disappear.

#	Author(s) & Year	Population / Sample	Music Condition(s)	Task Type (activity-demand level)	Outcome(s) Measured	Key Finding
26	*Yu & Chen (2024)*	94 students (61 design majors, 33 non-design), China	Cheerful music vs. melancholic music vs. silence	Novel / high-load (Attention Network Test)	Alerting, orienting, and executive-control attention networks	Among design majors, alerting and orienting attention were better in silence than under either music condition; executive control did not differ. Creative visual workers appear particularly susceptible to distraction by background music.
27	*Ba & Hu (2025)*	33 graduate students, mean age 26.5; native Chinese speakers reading in English; within-subjects (Asia-Pacific)	Background music varied by tempo (fast/slow) and mode (major/minor); five audio pieces	Novel / high-load (reading comprehension)	Reading comprehension; emotion via self-report, facial expression, and physiology (heart rate, skin temperature)	Tempo and mode did not directly affect comprehension, but music-induced emotions (valence, body temperature, heart rate) did, an effect fully mediated by emotion. Supports the arousal–mood hypothesis.
28	*Keeler, Puranik, Wang & Yin (2025)*	Online experiment plus 3-week experience-sampling field study of service employees (USA)	Workplace music fit vs. misfit (music matching vs. not matching the employee's momentary need)	Mixed (real-world service work)	Positive affect; cognitive depletion; citizenship and counterproductive behaviours; stimulus-screening ability (moderator)	Music misfit lowered positive affect and increased cognitive depletion, with effects stronger for employees lower in stimulus-screening ability. Locates the effect in the fit between music and listener, not the music alone.

#	Author(s) & Year	Population / Sample	Music Condition(s)	Task Type (activity-demand level)	Outcome(s) Measured	Key Finding
29	*Jäncke & Sandmann (2010)*	75 participants, randomly assigned to five groups (Switzerland)	Background music varied by tempo and consonance; noise as control	Novel / high-load (verbal learning)	Number of words learned; cortical activation (EEG alpha-band)	No effect of background music on verbal learning in either direction, though EEG showed distinct cortical activation across conditions. Authors suggest compensatory neural mechanisms maintain performance.
30	*Que & Hu (2025)*	52 non-native English speakers (ESL), aged 18–35 (Hong Kong)	Self-selected preferred background music vs. silence during reading	Novel / high-load (reading comprehension)	Reading comprehension; eye movements; EEG; heart rate; working memory, multitasking, proficiency, and listening habit (moderators)	No overall effect of music on comprehension or physiology, but individual traits moderated strongly: lower working memory and multitasking worsened outcomes under music; habituated listeners processed more efficiently. Lyrics predicted lower comprehension; faster tempo predicted more efficient processing.

Note. The activity-demand level follows the framework developed in Chapter 2. Routine activities are repetitive and low in cognitive load, on which music tends to support or not harm performance; familiar but effortful activities are those for which the listener already possesses the relevant knowledge or skill but which must still be performed effortfully, on which the effect of music is conditional; novel, high-load activities are knowledge-demanding (e.g., reading comprehension, verbal memory), on which music more often impairs performance. Studies employing sets of tasks or self-report that span more than one level are classified as mixed. Borderline cases were classified according to the cognitive demand of the primary task rather than the setting in which it occurred; where a study spanned more than one level, it was assigned to the category reflecting its dominant demand, or recorded as mixed. n = sample size; bpm = beats per minute; L1 = first language; L2 = second language; EFL = English as a foreign language; fNIRS = functional near-infrared spectroscopy; CPT = continuous performance test; OSpan = operation span; RT = reaction time.

Four systematic reviews/meta-analyses (Kämpfe et al., 2011; Cheah et al., 2022; de la Mora Velasco et al., 2023; Mendes et al., 2021) are not charted above as primary studies; they are discussed as higher-level framing evidence in Chapters 1, 4, and 5. Kämpfe et al. (2011) explicitly distinguishes task types, providing meta-analytic support for the activity-demand distinction. One further study (Jaušovec et al., 2006) is cited in Chapter 2 as contextual evidence on the Mozart effect but is not charted, because its priming paradigm involves listening before rather than during the task. One further empirical study (Cheah et al., 2025) is discussed in Sections 4.6 and 5.7 as complementary survey evidence on learners' reasons for studying with music but is not included in the charted set.

Although the activity-demand categories provide a useful structure for synthesis, some studies included tasks that overlapped more than one category. In such cases, classification was based on the dominant cognitive demands of the primary task, with borderline cases interpreted conservatively and recorded as mixed where no single level clearly applied.

4.2 Numerical Description of the Included Studies

A total of 30 studies met the eligibility criteria and were charted; they are summarised in full in Table 4.1. The distribution of the studies across the dimensions recorded during charting is described below and summarised in Table 4.2. Four reviews, treated as higher-level evidence rather than charted primary studies, are discussed separately in Section 4.7 and in the discussion that follows (Cheah et al., 2022; de la Mora Velasco et al., 2023; Kämpfe et al., 2011; Mendes et al., 2021).

4.2.1 Year of Publication

The included studies span the period from 2010 to 2026, although the eligibility window extended to work published in or after 2000, no eligible study predated 2010, and seminal earlier works are reserved for contextual discussion. The body of evidence is weighted towards the most recent decade, with a marked concentration of studies published from 2021 onward. This temporal pattern indicates that empirical interest in the relationship between music and cognitive performance has intensified rather than diminished in recent years, which itself strengthens the rationale for a current mapping of the field.

4.2.2 Population and Setting

The charted studies examined non-clinical populations across the age range specified in the eligibility criteria, encompassing secondary-school adolescents (e.g., Anderson & Fuller, 2010; Dobbs et al., 2011), university students (e.g., Avila et al., 2012; Souza & Barbosa, 2023; Sun et al., 2024; Su et al., 2023; Taheri et al., 2022), and healthy adults including older adults (Ferreri et al., 2013). Student and university samples predominate, consistent with the prominence of study and learning contexts in this literature. A subset of studies was conducted in workplace or work-analogue settings (Huang & Shih, 2011; Shih et al., 2012; Taheri et al., 2022), reflecting the deliberately broad scope adopted here, which treats occupational and educational activities as relevant according to their cognitive demand rather than their setting. Settings ranged from controlled laboratory environments through classroom and naturalistic study conditions to simulated work tasks, and one study drew on self-reported study-music habits rather than an experimental manipulation (Goltz & Sadakata, 2021). Several studies were conducted outside the Anglophone world, including in Portugal, Sweden, Iran, China, Vietnam, while Moreno and Woodruff (2024) contributes Canadian data. Together these lend the evidence base a degree of cross-cultural and cross-national range.

4.2.3 Type of Music

The music conditions examined varied widely, and the contrasts a study chose are themselves informative about the question it sought to answer. Several studies contrasted music with silence, while others compared categories of music against one another. Recurring contrasts included vocal or lyrical music versus instrumental music (Avila et al., 2012; Souza & Barbosa, 2023; Sun et al., 2024), fast versus slow tempo (Su et al., 2023; Lin et al., 2023; Moreno & Woodruff, 2024), and preferred versus non-preferred or self-selected music (Johansson et al., 2012; Kiss & Linnell, 2021; Vigl et al., 2023). The genres employed ranged from classical and instrumental pieces, including tempo-manipulated arrangements of Mozart's Sonata K.448 (Lin et al., 2023; Moreno & Woodruff, 2024) and instrumental classical music (Taheri et al., 2022; Su et al., 2023), to lo-fi hip-hop (Souza & Barbosa, 2023) and popular vocal music in several languages. The lyrical content of the music, in particular whether lyrics were present and in which language, emerged as a decisive dimension in the reading-comprehension studies and is examined in detail below.

4.2.4 Activity-Demand Level

When classified according to the activity-demand framework, the charted studies divided across the three levels and a mixed category. The largest group, fourteen studies, examined novel, high-load activities, most commonly reading comprehension and demanding verbal tasks, in which music competes for the limited resources the activity requires. A second group of five studies examined routine, low-load activities, such as sustained-attention and vigilance tasks and repetitive work, in which music may raise arousal toward an optimal level. A third group of five studies examined familiar but effortful activities, such as reading in a known language or holding material in working memory, for which the listener already possesses the relevant skill but must still perform the activity. The remaining six studies employed mixed sets of tasks or self-report spanning more than one level (Taheri et al., 2022; Vigl et al., 2023; Goltz & Sadakata, 2021; Cooke et al., 2026; Scott et al., 2025; Keeler et al., 2025). This distribution is itself informative: the literature has more often probed the conditions under which music harms demanding cognitive work than those under which it helps routine work, and the intermediate band, although theoretically important, is comparatively under-represented.

4.2.5 Outcomes Measured

The outcomes measured map onto the four themes of the research question. Reading comprehension was the single most frequently studied outcome, examined in ten studies, followed by memory and recall, including working memory, verbal encoding, and recognition, in six studies, and sustained attention or concentration in seven studies. A small number examined abstract reasoning or general ability (Dobbs et al., 2011; Avila et al., 2012). Motivation was measured least often, and where it appeared it was typically examined alongside mood and engagement rather than in isolation (Vigl et al., 2023; Goltz & Sadakata, 2021). Several studies also recorded individual-difference moderators, in particular working-memory capacity, personality in terms of introversion and extraversion, and music preference or listening habit, which recur throughout the narrative below. Because several studies measured more than one outcome, these categories overlap, and the counts are therefore indicative rather than mutually exclusive.

Table 4.2

Distribution of the Charted Studies by Activity-Demand Level and Primary Outcome

Category	Studies (n)	Approx. % of 30
Activity-demand level		
Routine (low-load)	5	17%
Familiar but effortful (intermediate)	5	17%
Novel / high-load	14	47%
Mixed / spans levels	6	20%
Primary outcome (overlapping)		
Reading comprehension	10	–
Memory and recall	6	–
Attention / concentration	7	–
Reasoning / general ability	2	–
Motivation / mood / learning	5	–

Note. Activity-demand levels are mutually exclusive and sum to the full set of charted studies. Outcome categories overlap, because several studies measured more than one outcome; the outcome counts are therefore indicative and do not sum to the total.

Taken together, these descriptive characteristics indicate that research has concentrated primarily on university students performing laboratory-based cognitive tasks, although the recent addition of large-scale, multi-study workplace evidence (Scott et al., 2025) means the occupational literature is better represented than in earlier syntheses. Comparatively fewer studies examined younger learners or longer-term educational outcomes. The conclusions of this review are therefore strongest for higher-education contexts and controlled experimental settings, with workplace conclusions resting on a smaller, if increasingly robust, evidence base.

4.3 Memory and Recall

The studies addressing memory present a divided picture that aligns closely with the activity-demand framework, and the division falls along the verbal content of the music rather than its mere presence. Where the memory activity was demanding and verbal, music carrying lyrics tended to impair performance, as though the verbal content of the music drew on the same processing the activity itself required. The clearest demonstration is Souza and Barbosa (2023), a methodologically strong within-subjects study of psychology students at the University of Porto, who completed verbal recall, visual recall, reading, and arithmetic tasks (with task-specific samples ranging from 106 to 123 participants) under three conditions: silence, instrumental lo-fi hip-hop, and Portuguese pop music with lyrics. Music with lyrics produced a small but credible impairment of verbal memory, visual memory, and reading ($d \approx -0.3$), while the effect on arithmetic was not credible. Instrumental music produced no credible effect in either direction. The study's distinctive contribution is metacognitive: participants correctly sensed that lyrics impaired them yet tended to misperceive instrumental music as beneficial when it was objectively neutral, a dissociation between felt and actual benefit that recurs as a theme later in the discussion. Avila et al. (2012) converge on the same conclusion from a different design, locating the interference of vocal music specifically in verbal processing rather than in numerical or logical tasks.

Individual differences further clarified this pattern. Christopher and Shelton (2017) found that the detrimental effect of background music on a demanding activity fell most heavily on individuals with lower working-memory capacity, while those with greater capacity were less affected. This finding fits naturally within a limited-resources account, since participants with more available capacity have more to spare for an additional auditory load, and it positions working-memory capacity as a moderator that determines the extent to which music disrupts performance. Nguyen and Grahn (2017), across three experiments with healthy adults, added a qualification concerning the music's own properties: low-arousal music supported recall and recognition more than high-arousal music, consistent with the inverted-U relationship between arousal and performance and with the principle that overly stimulating music can push a listener past the optimal point even on a memory activity.

However, two studies complicate this generally cautionary pattern, both involving instrumental music on demanding activities. Ferreri et al. (2013), using functional near-infrared spectroscopy with healthy older adults, found that background music during verbal encoding reduced prefrontal effort while improving memory performance, a pattern the authors summarised as "less effort, better results." Taheri et al. (2022) reported that classical instrumental music significantly improved working-memory performance on a two-back task, even though working memory is a demanding activity. Together, these two studies suggest that the impairment associated with demanding verbal activities is driven specifically by competing verbal or semantic content, that is, by lyrics, rather than by music as such, and that instrumental music can in some circumstances support even demanding memory activities.

Set against these findings is a well-controlled null result. Jäncke and Sandmann (2010) examined verbal learning in 75 participants under background music that varied in tempo and consonance, with noise as a control, while recording brain activity. Background music had no effect on the number of words learned, in either direction, yet the electroencephalographic data showed that the different acoustic conditions produced distinct patterns of cortical activation. The authors suggested that when background music draws additional attention, verbal learning may be held steady through compensatory neural effort. The study is a useful reminder that the absence of a behavioural effect does not mean the absence of any cognitive consequence, and that a null result at the level of performance can conceal real differences in how hard the brain is working.

4.4 Concentration and Attention

The evidence on concentration and sustained attention provides the clearest support for the proposition that music can support, or at least not harm, performance on routine, low-demand activities. Kiss and Linnell (2021) found that preferred background music increased task focus and reduced mind-wandering during a low-demand sustained-attention task in a sample of 40 students. In a subsequent and more elaborate study, Kiss and Linnell (2024) replicated and extended this result across two experiments with adults habituated to studying with music, showing that music decreased mind-wandering, increased task focus, and speeded reaction times, with the effects statistically mediated by mood and arousal. The identification of mediation is theoretically significant, because it specifies the mechanism, an arousal- and mood-based lift, by which music is thought to benefit under-stimulating activities, and it provides direct rather than merely inferential support for the arousal-mood hypothesis

The workplace and work-analogue studies are consistent with this pattern while qualifying it. Huang and Shih (2011) found that background music affected worker concentration, with the direction of the effect depending on music type and individual preference, and Shih et al. (2012) reported that attention performance varied with the type of music and the duration of exposure. These studies indicate that the benefit of music for routine activities is conditional rather than guaranteed: it depends on the music being acceptable to the listener and on the activity being under-stimulating. Lin et al. (2023), examining cognitive processing speed across motor, visuospatial, and linguistic tasks in Vietnamese undergraduates, add a further nuance from the tempo dimension: slow-tempo music worsened processing speed across all three tasks relative to no music, while fast-tempo music produced no speed benefit over silence, and, notably, arousal did not mediate the effect. This complicates any simple arousal-based reading of tempo and indicates that the relationship between tempo and routine performance is not yet well understood.

A direct counter-instance must also be recorded. Taheri et al. (2022), whose participants performed a continuous performance test of sustained attention alongside the working-memory and dexterity tasks noted above, found that background music had no significant effect on sustained attention, even as it improved working memory and motor-skill performance in the same participants. This dissociation within a single study is a useful corrective, since it indicates that the benefit of music for low-demand activities is not universal and that the effect of music can differ across activities performed by the same people.

Two recent studies extend the attention evidence in opposite settings. Scott et al. (2025) substantially strengthen the occupational literature, which is otherwise thinly represented. Across a two-week experience-sampling pilot of 108 employees, a laboratory study of 252 undergraduates completing proofreading tasks, and a three-week experience-sampling study of 247 employees with coworker performance ratings, the amount of time spent listening to music at work related to performance in an inverted-U pattern rather than a linear one, with the curvilinear cost of extended listening more pronounced among those who believed willpower to be a limited resource. Its finding of a curvilinear relationship provides direct empirical support for the inverted-U principle set out in Chapter 2. Yu and Chen (2024), by contrast, examined attention among design students, a group selected for highly creative visual thinking, using the Attention Network Test under cheerful music, melancholic music, and silence. Among the design majors, alerting and orienting attention were significantly better in silence than under either music condition, while executive control did not differ, leading the authors to conclude that those engaged in demanding creative work are particularly susceptible to distraction by background music.

Keeler et al. (2025) show that what matters for workplace music is not the music itself but how well it fits the listener. Across an online experiment and a three-week experience-sampling study of service employees, a mismatch between the music playing and the employee's momentary needs lowered positive affect and increased cognitive depletion, and these costs fell most heavily on employees less able to screen out irrelevant stimulation. The effect of workplace music thus depended not on the music alone but on whether it suited the individual, which fits the wider theme that listener characteristics shape whether music helps or hinders.

The balance of the evidence is therefore that music more often supports than harms attention on low-demand activities, but that the benefit is contingent on music type, tempo, listener preference, and the specific attentional demand involved.

4.5 Reading and Learning

Reading comprehension is the most thoroughly studied outcome in this literature, and it provides the strongest and most consistent evidence for impairment on novel, high-load activities, with some more mixed evidence concerning tempo. The recurring finding is that music containing lyrics disrupts reading comprehension. Anderson and Fuller (2010), in a large sample of 334 American seventh- and eighth-grade students, found that lyrical music reduced reading comprehension, with the largest reductions among those who reported preferring to study with music, a counter-intuitive result suggesting that habitual exposure does not necessarily protect learners from disruption and may accompany greater vulnerability. Sun et al. (2024) added further detail to this pattern. Examining 90 Chinese college students reading in their first and second languages, and distinguishing habitual listeners from non-listeners, they found that pop music with lyrics reduced comprehension relative to silence, that lyrics in the same language as the text were the most disruptive, and that native-language lyrics were the most disruptive of all. They interpreted these results through the duplex-mechanism account of auditory distraction, under which semantically meaningful lyrics interfere specifically with the semantic processing that reading itself requires.

This interference-based explanation is reinforced by the role of individual differences. Sun et al. (2024) found that habitual listeners were less impaired than non-listeners, consistent with the view that experience and stronger inhibitory control can reduce, though not eliminate, the interference. The convergence of these reading studies on a lyrics-driven, semantic-interference explanation is one of the clearer signals in the charted evidence, and it is precisely the kind of graded, mechanism-specific finding that distinguishes a clearer pattern from a merely contradictory one.

The mixed evidence concerns tempo and preference, and it complicates any simple conclusion that music harms reading. Su et al. (2023), using eye-tracking with Chinese English-majors reading in English, found that fast-tempo instrumental music was associated with faster reading than slow-tempo music, that the tempo effect was more pronounced on easy texts than difficult ones, and that participants with a high preference for studying with music outperformed those with a low preference, who were most hindered on difficult texts under slow-tempo music. By contrast, Moreno and Woodruff (2024), studying 74 first-year undergraduates with a multimodal design that recorded facial-emotion expression and skin-conductance alongside comprehension, reported that fast tempo lowered comprehension and raised markers of fear and physiological arousal relative to slow tempo, which produced the highest scores. The apparent contradiction, fast tempo helping in one study and harming in another, points to genuine moderating factors rather than to error. It suggests that tempo interacts with task difficulty, listener preference, and the specific outcome measured, in particular whether the dependent variable is reading speed or comprehension accuracy.

What unifies the reading evidence is therefore not a single direction of effect, but a clear message that lyrical, semantically rich music is reliably disruptive to comprehension, while the effects of instrumental music depend on tempo, text difficulty, and preference. Johansson et al. (2012), using eye-movement measures, similarly found that music preference shaped how reading was processed, reinforcing the conclusion that the listener cannot be separated from the music when interpreting effects on reading.

Que et al. (2023) add a methodologically distinctive contribution by using self-selected rather than researcher-imposed music, addressing the concern that participants rarely hear music of their own choosing. Studying 100 non-native English speakers with eye-tracking, they found no difference in reading comprehension accuracy between the music and silence conditions, but the eye-movement data showed that music increased cognitive load specifically at post-lexical rather than lexical stages of processing, indicating that the disruption was not uniform across the reading process. Higher English proficiency and more frequent habitual use of study music were each associated with reduced load under music, consistent with the habituation pattern seen in Sun et al. (2024), whereas higher working-memory capacity was associated with greater invested effort rather than less, a finding that complicates the straightforward expectation that greater capacity is uniformly protective.

Ba and Hu (2025) point to why tempo effects on reading have proved so variable. In a within-subjects study of 33 graduate students reading in a second language while listening to music that varied in tempo and mode, they found that neither property affected comprehension directly, but that the emotion the music induced, measured through self-report and physiological signals, did. A musical feature such as tempo therefore appears to reach reading performance through the listener's emotional response rather than by acting on comprehension in its own right, which helps explain why studies manipulating tempo alone have reached inconsistent conclusions.

Que and Hu (2025) combined eye-tracking, electroencephalography, and heart-rate measures while 52 non-native English speakers read with their own preferred music or in silence. Background music had no significant overall effect on comprehension accuracy or on the physiological measures, but individual characteristics moderated the effect: readers with lower working-memory capacity and poorer multitasking ability showed longer reading times and greater cognitive load under music, whereas those accustomed to studying with music processed text more efficiently. Considered separately, the presence of lyrics was associated with lower comprehension accuracy and more disrupted word-level processing, while faster tempo was associated with more efficient text processing. The absence of an overall effect alongside these moderating influences is consistent with the pattern seen elsewhere here, in which the effect of music on reading depends on the listener and the music rather than following a single direction.

Beyond comprehension narrowly defined, Vigl et al. (2023) examined self-selected music in a classroom learning context and found that positive, high-energy music improved mood, motivation, and learning, with effects varying across individuals. Because this study spans learning, motivation, and concentration in a naturalistic setting, it sits at the boundary of several themes and is discussed further under motivation below.

4.6 Motivation

Motivation was the least frequently isolated outcome in the charted studies, and where it appeared it was generally examined together with mood and engagement rather than as a discrete construct. The most direct evidence comes from Vigl et al. (2023), who found that self-selected, high-energy classroom music improved students' mood, motivation, and concentration and supported learning, though with substantial individual variation. The survey study by Goltz and Sadakata (2021) complements this by mapping how and why people choose to study with music in real-world settings, indicating that learners frequently use music deliberately to regulate their motivation and focus, whatever its measurable effect on performance. Beyond the charted studies, Cheah et al. (2025) extended this picture using a combination of retrospective survey and six-week experience-sampling data and found that students most often turned to study-music for affect regulation and concentration, to fill silence, block out noise, and sustain focus, rather than in the expectation of a direct performance gain. Self-reported mood did not measurably rise over study sessions, which led the authors to suggest that music serves to maintain a working affective state rather than to elevate it.

Cooke et al. (2026), surveying 226 university students on their music-listening habits while reading, reinforce this motivational picture. Around half the sample (54%) reported listening to music while reading for study and half (46%) avoided it, and the most commonly cited reasons for listening were to help focus (57%), to mask external noise (50%), and to increase motivation (49%), with open-ended responses pointing additionally to emotion and stress management. The most frequently chosen music was non-lyrical and slow in tempo, consistent with a deliberate strategy of regulating arousal to suit the activity. These findings locate motivation and affect regulation, rather than an expected gain in measured performance, at the centre of why learners choose to study with music.

Sun (2025) provides rare experimental evidence bearing on motivation as an outcome. In a two-phase study of 428 undergraduates comparing control, high-arousal, low-arousal, and Mozart K.448 conditions on a working-memory task, structured instrumental music supported performance indirectly through flow and work engagement, whereas high-arousal music was detrimental. A 30-day follow-up showed that the benefit of the structured music attenuated substantially with familiarity but did not disappear, echoing the role of habituation in the charted evidence. That the motivational benefit operated through flow and engagement, rather than directly on performance, is consistent with the mood- and arousal-mediated pathway proposed by the arousal–mood hypothesis.

Two observations follow. First, the motivational and affective benefits of music appear to operate through the same arousal- and mood-based pathway implicated in the attention findings, which suggests a degree of theoretical coherence across the motivation and concentration themes. Second, although learners consistently cite motivation and focus as their main reasons for studying with music, few studies have measured motivation as a discrete outcome rather than folding it into mood or engagement. This constitutes a genuine gap in the evidence base, and one revisited in the Discussion.

4.7 Summary of the Mapped Evidence

When read through the activity-demand framework, the charted evidence shows a broadly coherent but not uniform pattern. At the routine end of the spectrum, under-stimulating activities such as sustained attention, vigilance, and repetitive work were the contexts in which music most often supported performance, mainly by increasing task focus and reducing mind-wandering through mood and arousal. At the novel, high-demand end, demanding verbal activities, particularly reading comprehension and effortful verbal memory, were the contexts in which music most often impaired performance, especially when the music included lyrics. The intermediate familiar-but-effortful category, including activities such as reading in a known language and holding material in working memory, showed the most variable effects, which is consistent with its position between these two mechanisms. This broad division is consistent with the larger body of evidence beyond the charted studies: the systematic review by Cheah et al. (2022), spanning 154 experiments, likewise found music's effects to be small, frequently absent, and concentrated as impairment in the memory and language domains, with lyrical music more disruptive than instrumental and harder activities more affected than easy ones.

Three qualifications prevent this pattern from being treated as a fixed rule. First, on demanding activities, the disruptive factor appears to be lyrical content rather than music itself, since instrumental music was often neutral and was occasionally associated with better performance.

Second, the activity-demand framework should be understood as an organising tool rather than a strict predictor. For example, Taheri et al. (2022) found that instrumental music improved working-memory performance but had no significant effect on sustained attention, while the tempo studies produced different findings depending on text difficulty, listener preference, and whether speed or accuracy was measured. Third, listener characteristics were important across the evidence. Working-memory capacity, personality, music preference, and listening habit often shaped not only the size of an effect but also its direction. The discussion that follows considers how these patterns relate to the wider meta-analytic evidence and discusses their implications for educational and occupational practice.

A final observation concerns the differing weight the charted studies can bear. In keeping with scoping-review methodology, no formal quality appraisal was conducted, but the studies are not uniform in their evidential strength, and this should temper how firmly individual findings are read. Some rest on robust designs that invite greater confidence: Scott et al. (2025), for instance, combined multiple large samples with objective and coworker-rated performance, and Souza and Barbosa (2023) used a well-powered within-subjects design with task-specific samples above one hundred participants. Others rest on smaller or single-sample studies whose findings should be treated more cautiously, such as the 60-participant eye-movement study of Su et al. (2023) or the small work-analogue samples in the occupational literature. Where several studies of differing design converge, as they do on the disruptive effect of lyrics on reading, the conclusion can be held with reasonable confidence; where a single small study stands alone, as with some of the tempo and workplace findings, the mapping records the result but does not lean heavily upon it. The pattern across reading, memory, attention, and motivation is less four separate stories than one activity-demand-shaped pattern viewed through four different outcome measures.

Chapter 5: Discussion

5.1 Purpose of the Review and Principal Findings

The purpose of this scoping review was to map empirical evidence on the effects of music on memory, learning, motivation, and concentration during study and work-related cognitive tasks, and to examine how those effects are shaped by task demand, musical characteristics, and individual differences. In keeping with the purpose of a scoping review, no single verdict is offered on whether music improves cognitive performance. Instead, the patterns that emerged from the charted evidence are interpreted, related to the theoretical activity-demand framework, and considered for what they imply for educational and occupational practice and for future research, alongside the areas where the evidence remains mixed.

Three main findings emerged from the evidence mapped here. First, the direction of music's effect varied with the cognitive demand of the concurrent activity: music tended to support performance, or at least not disrupt it, during routine, low-load activities; produced mixed and conditional effects during familiar but effortful activities; and was more likely to impair performance during novel, high-load activities. Second, within the high-load category, impairment appeared to be linked especially to lyrical content rather than to music itself. Instrumental music was frequently neutral and was occasionally associated with better performance, even during demanding activities. Third, individual differences in working-memory capacity, personality, music preference, and listening habit moderated these effects, in some cases shaping both their strength and direction.

The theoretical perspectives introduced earlier do not compete to explain these findings; they address different parts of the same picture. Cognitive load theory accounts for why interference becomes more likely as an activity's demands increase and the resources available to absorb an additional auditory stream shrink accordingly. The arousal-mood hypothesis accounts for the opposite pattern at the other end of the spectrum, explaining why music can support performance precisely when an activity leaves resources spare: under those conditions, a lift in arousal or mood is a net gain rather than a competing demand. Individual differences determine where, for a given listener, that balance actually falls. Working-memory capacity, personality, and listening habit each shift the point at which competition for resources begins to outweigh any motivational benefit.

5.2 The Activity-Demand Pattern: From Routine to Novel Tasks

The most consistent pattern in the charted evidence is the dependence of music's effects on the cognitive demand of the activity, and it aligns closely with theory: it is precisely what the joint operation of the arousal–mood mechanism and the limited-capacity mechanism predicts. On routine, under-stimulating activities, the main risk for performance is disengagement, and music's capacity to raise arousal and improve mood addresses that risk directly. This is seen most clearly in the sustained-attention studies of Kiss and Linnell (2021, 2024), in which preferred music reduced mind-wandering and improved task focus, with mood and arousal statistically mediating the benefit. The identification of mediation is especially important: it raises the arousal–mood hypothesis from a plausible interpretation to one with direct empirical support within the charted evidence. The workplace studies (Huang & Shih, 2011; Shih et al., 2012) point the same way while adding an important qualification: the benefit is conditional on the music being acceptable to the listener and on the activity being undemanding.

On novel, high-load activities the pattern reverses, and the reading-comprehension literature supplies the clearest demonstration. Lyrical music reliably impaired comprehension across populations and designs (Anderson & Fuller, 2010; Sun et al., 2024; Souza & Barbosa, 2023), and the form of the impairment points to a specific mechanism. In Sun et al. (2024), lyrics in the same language as the text were more disruptive than lyrics in another language, and native-language lyrics were the most disruptive of all. This graded specificity is what the interference-by-process component of the duplex-mechanism account predicts (Hughes, 2014): disruption scales with the overlap between the processing the sound recruits and the processing the activity requires. The convergence between the meta-analytic evidence (Kämpfe et al., 2011), the theoretical framework, and the charted primary studies on this point is one of the more consistent conclusions to emerge from the mapping.

Between these poles lie familiar but effortful activities, for which the listener already possesses the relevant knowledge or skill but must still perform the activity, and it is here that the evidence is most mixed. The clearest illustrations come from studies that placed instrumental music against a moderately demanding activity. Taheri et al. (2022) found that classical instrumental music improved working-memory performance on an n-back task, a demanding outcome, while leaving sustained attention, a low-demand outcome, unaffected; the same study therefore shows both facilitation and no effect within a single sample. Ferreri et al. (2013) found that music reduced prefrontal effort while improving verbal encoding in older adults. The tempo findings for reading point in opposite directions across studies, with fast-tempo instrumental music associated with faster reading in Su et al. (2023) but with poorer comprehension and higher stress in Moreno and Woodruff (2024). These results do not overturn the framework. Rather, they specify its boundary conditions: the demand-based prediction holds most firmly when the music contains competing verbal content and when the outcome measured is comprehension accuracy rather than reading speed, and it is at its least determinate in the intermediate band, where the characteristics of the music and the listener decide the direction of the effect. Understood this way, the variability in the intermediate category shows why this category is useful for interpreting mixed findings.

5.3 Lyrical Content as the Main Source of Interference

The most practically useful refinement to emerge from the mapping is that the question of whether music impairs demanding cognitive work is better replaced by the question of whether the music contains semantic content that competes with the activity. Across the charted studies, the contrast between lyrical and instrumental music appeared more informative than the contrast between music and silence. Souza and Barbosa (2023) found that lyrics hindered verbal memory, visual memory, and reading, while instrumental music was broadly neutral; Avila et al. (2012) located the interference of vocal music specifically in verbal processing; and the language gradient in Sun et al. (2024) ties the disruption to semantic processing rather than to acoustic stimulation as such. The theoretical implication is that the limited-capacity competition described earlier operates not over an undifferentiated pool of attention but over process-specific resources. The practical implication is that guidance to learners should target lyrics rather than music in general.

This refinement also helps to reconcile the apparent conflict between the impairment literature and the facilitation findings. If interference is process-specific, instrumental music leaves the verbal channel largely free, and its arousal–mood benefits can then operate even during demanding work, which is plausibly what was observed in Taheri et al. (2022) and Ferreri et al. (2013). On this reading, the cases that sit awkwardly with a simple demand-based prediction may be better understood as situations in which the interference mechanism was reduced or absent, because no lyrics were present, while the facilitation mechanism remained active.

5.4 The Pervasive Role of Individual Differences

The third principal finding is that individual differences are not peripheral but important moderators of music's effects. Working-memory capacity buffers the cost of music on demanding activities (Christopher & Shelton, 2017), consistent with the resource account, since listeners with greater available capacity have more to spare. Personality moderates in the direction Eysenck's arousal theory predicts, with introverts, who are already closer to optimal arousal, more readily tipped into impairment than extraverts (Dobbs et al., 2011), and several charted studies finding stronger benefits among extraverts (e.g., Taheri et al., 2022). Music preference and listening habit moderate through both mechanisms at once: preferred music delivers a more reliable arousal-mood lift (Kiss & Linnell, 2021), and habitual listeners appear better able to resist distraction, as in the listener and non-listener contrast of Sun et al. (2024), in which students accustomed to studying with music were substantially less impaired by it. The practical consequence is that general prescriptions, whether to study in silence or that music aids focus, are both untenable as universal claims. The defensible guidance is conditional on the listener as well as on the activity and the music.

The survey evidence of Cooke et al. (2026) adds a useful qualification to the working-memory account. Contrary to expectation, they found no association between working-memory capacity, measured by an operation-span task, and the decision to listen to or avoid music while reading, and no association for trait mind-wandering either; instead, the variable that distinguished listeners from avoiders was music engagement, with more musically engaged students both more likely to listen and more likely to judge the music helpful. This does not contradict the experimental finding of Christopher and Shelton (2017) that working-memory capacity buffers the cost of music on demanding activities, since the two studies measure different things: Christopher and Shelton tested performance under music, whereas Cooke et al. measured self-reported listening habits rather than comprehension. Read together, they suggest that working-memory capacity may shape how much music actually disrupts performance without necessarily shaping whether a learner chooses to listen, a choice that appears to be driven more by engagement and preference, and the dissociation is itself consistent with the recurring theme that felt benefit and measured benefit do not always coincide.

This divergence between perceived and measured benefit deserves emphasis, because it recurs across the evidence and carries a specific implication. Souza and Barbosa (2023) found that participants correctly sensed the cost of lyrics but misperceived instrumental music as helpful when it was objectively neutral, and Cooke et al. (2026) found that the perception of music as helpful tracked music engagement rather than any measured capacity to resist distraction. These results suggest that a listener's subjective sense of how music affects their performance is an unreliable guide to its actual effect. This bears directly on the idea that learners can, through experience, learn to manage music as a study aid. Effective self-regulation depends on accurate feedback, and if learners systematically misjudge when music is helping and when it is not, their capacity to manage it through experience is correspondingly limited. The evidence therefore qualifies the more optimistic reading of habituation: experience may reduce the disruptive effect of music, as the listener and non-listener contrast in Sun et al. (2024) suggests, but it does not appear to make learners accurate judges of that effect.

The same moderating logic reaches into the workplace. Keeler et al. (2025) found that the costs of ill-fitting background music fell most heavily on employees least able to screen out irrelevant stimulation, a trait close in spirit to the working-memory and distractibility differences discussed above, and the multimodal study by Que and Hu (2025) points the same way, finding no overall effect of music on reading yet clear moderation by working-memory capacity, multitasking ability, and listening habit. Whether music helps or harms therefore depends not only on the music, but on who is listening, in the workplace as in the classroom

5.5 Relationship to the Wider Review-Level Evidence

The present findings are broadly consistent with previous reviews in indicating that music does not exert uniformly positive or negative effects on cognition. Whereas earlier reviews have generally synthesised outcomes according to cognitive domain or overall effect size, however, the analysis here suggests that activity demand offers a useful organising principle for explaining why apparently contradictory findings emerge across studies.

The patterns mapped here align with the higher-level evidence identified during the search. Kämpfe et al. (2011) found, meta-analytically, that the global effect of background music on adults is close to null but conceals systematic demand-dependent sub-effects, a conclusion the present mapping reproduces at the level of primary studies. Cheah et al. (2022), reviewing task, music, and population factors across a large body of experiments, similarly emphasised that effects are conditional on the interaction of all three, and de la Mora Velasco et al. (2023) reached a comparable conclusion for learners specifically. The agreement between the present scoping map and these independent syntheses strengthens confidence that the activity-demand pattern, the lyrics refinement, and the centrality of individual differences are properties of the literature itself rather than artefacts of study selection.

5.6 Implications for Educational and Occupational Practice

Although a scoping review does not weigh evidence toward clinical-style recommendations, the consistency of the mapped patterns supports several cautious, conditional implications. The educational implications are the most direct, since studying is where the evidence is densest and where learners most often exercise a personal choice about music. The evidence suggests avoiding music with lyrics, particularly familiar music with native-language lyrics, during reading, comprehension, and other verbally demanding study activities, since this is the configuration in which impairment is most reliable. Where students wish to study with music, instrumental music at moderate arousal levels is the configuration least likely to harm and most likely to deliver the mood and motivation benefits that learners themselves report seeking (Goltz & Sadakata, 2021; Vigl et al., 2023). Students who have no habit of studying with music should be aware that they are likely to be more, not less, vulnerable to its costs than habitual listeners are (Sun et al., 2024).

For educators, the classroom evidence suggests that music can be a legitimate tool for managing mood, motivation, and engagement, particularly during routine, procedural, or low-load activities, but that it is best withdrawn during reading, composition, and other high-load verbal work. The workplace evidence points in a symmetrical direction and corroborates the educational picture across a different setting: background music is most defensible for repetitive, monotonous work, where it can sustain arousal and reduce disengagement (Huang & Shih, 2011; Shih et al., 2012; Taheri et al., 2022), and least defensible for knowledge work involving sustained reading and writing. That the same activity-demand pattern appears in both classroom and workplace settings is itself instructive: what governs the effect is the demand of the activity, not the setting in which it is performed. In both contexts, the moderating force of preference and habit makes individual control over whether and what music plays more defensible than any uniform policy.

5.7 Gaps in the Evidence and Directions for Future Research

The mapping exercise identified several gaps that future research should address. First, motivation is markedly under-studied as a discrete outcome. Although learners' own reasons for studying with music are substantially motivational (Goltz & Sadakata, 2021; Cheah et al., 2025), few studies isolate motivation from mood and engagement, and fewer still measure it with the rigour routinely applied to memory and comprehension. Second, the tempo literature is unresolved: the opposing findings of Su et al. (2023) and Moreno and Woodruff (2024) indicate that tempo interacts with task difficulty, listener preference, and the outcome measured, and systematic designs that cross these factors are needed. Third, the workplace evidence remains limited relative to the educational evidence, despite the prevalence of music listening at work; studies using ecologically valid occupational activities would considerably strengthen this area of the literature.

Fourth, the semantic-interference explanation that best accounts for the reading findings invites finer-grained tests. Research is beginning to ask whether interference operates on a graded continuum of language familiarity, from foreign-language through second-language to native-language lyrics, rather than as a binary presence or absence of intelligible content. A registered report by Ding and Choi (2025) is designed to test exactly this gradation, comparing native-, second-, and foreign-language lyrics in both first- and second-language reading, and its results, once available, should inform the theoretical refinement of the duplex-mechanism account. Fifth, the field would benefit from greater standardisation, though of a particular kind. Because preference and familiarity emerged as decisive moderators, standardising the stimulus into uniformity, by imposing the same tempo, genre, and lyrical content on every participant, would strip out precisely the subjective dimensions found to matter, and would trade external validity for a control that misrepresents how people actually listen. What is needed instead is standardisation of measurement and reporting: consistent documentation of the objective properties of the music, such as tempo, volume, lyrical content, language, and familiarity, combined with systematic capture of listeners' subjective responses to it, in particular their preference for and liking of the music used. Standardising how these subjective and objective features are recorded, rather than standardising the music itself, would allow comparison across studies without discarding the individual differences that the evidence identifies as central.

5.8 Strengths and Limitations of the Review

The principal strengths of this review are its breadth and its transparency. The deliberately broad scope, which spans educational and occupational settings, child and adult populations, and the full range of music characteristics, allowed the activity-demand pattern to emerge across contexts in a way a narrower review could not have shown, and the methods adopted (the PRISMA-ScR framework, a priori eligibility criteria, and systematic charting) make the mapping reproducible. The use of an explicit theoretical lens, declared in advance and applied as a heuristic rather than a filter, allowed the evidence to be organised without being forced.

Several limitations must be acknowledged. First, in keeping with scoping-review methodology, no formal quality appraisal was undertaken. The mapped findings should therefore not be read as having been weighted by methodological quality, and results from small-sample studies should be interpreted with caution. Second, the restriction to English-language publications may have excluded relevant evidence, which is an important consideration in a field where several active research groups publish in other languages. Relatedly, because several high-yield databases were screened only to a fixed, relevance-ranked depth (the first 100 records for PsycINFO and Scopus, the first 50 for Google Scholar, and the first 30 for ERIC), eligible studies ranked below those cut-offs may have been missed. Backward citation searching, the manual journal search, and the fact that Google Scholar surfaced no eligible studies beyond those already identified all mitigate this risk, but they cannot eliminate it. Third, the review's reliance on published, peer-reviewed sources means that it may be affected by publication bias, particularly if studies showing striking benefits or costs of music are more likely to be published. Fourth, the activity-demand framework, while theoretically grounded, required judgement in borderline cases, particularly for activities in the familiar but effortful category, and a different classification of mixed-activity studies could shift the apparent strength of the pattern at its margins. More broadly, the framework itself is one interpretive lens among others that could have been applied to the same evidence; a synthesis organised instead around outcome type or musical characteristic might draw somewhat different conclusions from the same studies. Fifth, the review was conducted by a single researcher, so study selection and charting were performed by one reviewer rather than by two independent reviewers; although the eligibility criteria and charting form were defined in advance to reduce subjectivity, the absence of a second reviewer means the possibility of selection or extraction error cannot be fully excluded. Sixth, although the review was deliberately scoped to include both educational and occupational activities, the charted studies remain weighted towards educational contexts, and workplace evidence, while now considerably stronger than in earlier syntheses, still rests on fewer studies overall. Conclusions drawn about workplace performance should accordingly be read with somewhat more caution than those drawn about study contexts. These limitations do not diminish the value of the mapping, but they define the level of confidence with which its conclusions should be read.

Chapter 6: Conclusion

This thesis asked how, and under what conditions, music affects memory, learning, motivation, and concentration during study and work-related cognitive tasks, and how the demands of the activity, the characteristics of the music, and differences between individuals shape those effects. A scoping review of the empirical literature, conducted in accordance with the PRISMA-ScR framework and spanning educational and occupational contexts, supports a conditional interpretation of the evidence. Music can both support and impair cognitive performance, and the direction of its effect depends largely on the demand of the activity, the verbal content of the music, and the characteristics of the listener.

The first conclusion concerns the nature of the activity. The effects of background music vary systematically along a spectrum of cognitive demand. On routine, low-load activities, such as sustained attention, vigilance, and repetitive work, music is more likely to support than disrupt performance, by raising arousal, lifting mood, and reducing mind-wandering. On novel, high-load activities, above all reading comprehension and demanding verbal learning, music is more likely to impair performance, by competing for the limited processing resources the activity requires. Familiar but effortful activities, for which the necessary knowledge or skill is already possessed but the activity must still be performed, fall between these two positions, where the effect of music is conditional and least predictable. This graded pattern, predicted jointly by the arousal-mood hypothesis and cognitive load theory and supported by the meta-analysis of Kämpfe et al. (2011), held across both the educational and the workplace evidence charted here, indicating that the principle is linked to the cognitive demand of the activity rather than to the setting in which it is performed.

The second conclusion concerns the characteristics of the music. The main source of impairment appears to be the presence of lyrics rather than music as such. Across the charted studies, music with semantically meaningful lyrics, and particularly familiar music with native-language lyrics, was reliably disruptive to verbal activities, whereas instrumental music was frequently neutral and occasionally beneficial even during demanding work. The interference-by-process component of the duplex-mechanism account of auditory distraction (Hughes, 2014) captures the mechanism: disruption is greatest when the music recruits the same semantic processing the activity requires. To the practical question of which music is least disruptive to concentration, the evidence answers that instrumental music, at moderate arousal and preferably self-chosen, is the configuration least likely to harm, and that during verbally demanding work, silence remains the least risky condition.

The third conclusion concerns individual differences between listeners. These differences are central to understanding music's effects. Working-memory capacity, personality, music preference, and the habit of studying with music each moderated the mapped effects, in some cases shaping both their strength and direction. Because the same music on the same activity can support one person and hinder another, no uniform prescription is supported by the evidence; guidance should be conditional, and allowing individuals control over whether and what music plays is the recommendation the evidence best sustains. At the same time, learners' subjective sense that music helps them should be treated as useful but not sufficient evidence, since perceived benefit does not always correspond with measured performance.

These conclusions carry direct implications for educational practice. Learners are best advised to avoid music with lyrics during reading, writing, and memorisation, to prefer instrumental music at moderate arousal where they wish to study with music at all, and to recognise that those who do not habitually study with music are likely to be more vulnerable to its costs. Educators may treat music as a possible tool for sustaining mood, motivation, and engagement during routine or procedural work, while withdrawing it during high-load verbal activity. Because these implications follow the demand of the activity rather than the setting, they extend naturally from the classroom to the workplace. They are particularly relevant to self-regulated learning: students often decide independently whether to listen to music, what kind of music to choose, and when to use it, and such decisions are better guided by the demands of the specific activity being undertaken than by a general belief that music helps or harms.

Important questions remain open. Motivation, which is among the most common reasons learners give for studying with music, is the least studied outcome as a discrete construct. The effects of tempo remain unresolved, with well-conducted studies pointing in opposite directions. The workplace evidence remains limited relative to its practical importance. The semantic-interference explanation that best accounts for the reading findings invites finer-grained tests of how interference varies with the familiarity of the lyrics' language, which ongoing research is beginning to undertake. Together these gaps point towards a clear agenda for the next phase of research.

These conclusions should also be read in keeping with the nature of the study. As a scoping review, the thesis set out to map the range and pattern of the evidence rather than to estimate the size of music's effect, so its findings describe broad tendencies and conditions rather than precise predictions for any particular learner, activity, or listening situation.

Within those bounds, the contribution of this thesis is to show that a body of evidence long regarded as contradictory becomes substantially coherent once it is mapped broadly and read through the activity-demand framework. Whether music helps or hinders cognition cannot be answered in a single or general way, but the narrower and more useful questions, which activities, which music, and which listeners, can be. Drawing these strands together provides learners and educators with a stronger evidence base for decisions that are at present often made by intuition and indicates that how music affects studying and working is best understood not as a fixed property of music but as the joint product of the activity, the music, and the listener.

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Appendix A: Full Search Documentation

This appendix documents the search conducted for each source, including the exact string used, the date the search was run, and the number of records returned. Records were adapted to the syntax and controlled vocabulary of each database while preserving the same underlying Population, Concept, and Context elements. Where a database returned a very large number of records, results were sorted by relevance and screened to a fixed depth, as noted for each source below and as summarised in the PRISMA-ScR flow diagram (Figure 3.1).

A1. PubMed / MEDLINE

Search string:

("Music"[Mesh] OR "background music"[tiab] OR "music listening"[tiab] OR "instrumental music"[tiab]) AND ("Attention"[Mesh] OR "Memory"[Mesh] OR "Learning"[Mesh] OR "Motivation"[Mesh] OR concentration[tiab] OR "cognitive performance"[tiab] OR "task performance"[tiab] OR recall[tiab] OR "academic performance"[tiab]) AND (students[tiab] OR pupils[tiab] OR learners[tiab] OR children[tiab] OR adolescents[tiab] OR workers[tiab] OR employees[tiab] OR workplace[tiab] OR occupational[tiab])

Date searched: February 2026. *Records returned:* 541 (screened in full).

A2. PsycINFO (APA PsycNet)

Search string:

("background music" OR "music listening" OR "instrumental music") AND (concentration OR attention OR "cognitive performance" OR memory OR learning OR motivation) AND (students OR workers OR employees)

Filters applied: Peer Reviewed = Yes; Document Type = Academic Journal.
Date searched: March 2026. *Records returned:* 1,605 after filters; results were sorted by relevance and the first 100 were screened.

A3. ERIC

Search string:

background music AND (concentration OR attention OR cognitive performance OR memory OR learning OR motivation) AND (students OR workers OR employees)

Date searched: June 2026. *Records returned:* 769; results were sorted by relevance and the top 30 were screened.

A4. Scopus

Search string:

TITLE-ABS-KEY((“background music” OR “music listening” OR “instrumental music”) AND (concentration OR attention OR “cognitive performance” OR memory OR learning OR motivation) AND (students OR workers OR employees))

Date searched: April 2026. *Records returned:* 17,273; results were sorted by relevance and the first 100 were screened.

A5. Web of Science

Search string:

TS=((“background music” OR “music listening” OR “instrumental music”) AND (concentration OR attention OR “cognitive performance” OR memory OR learning OR motivation) AND (students OR workers OR employees))

Date searched: June 2026. *Records returned:* 12 (screened in full).

A6. Google Scholar

Search string:

“background music” (“reading comprehension” OR “cognitive performance” OR concentration OR memory) (students OR workers)

Date searched: May 2026. *Records returned:* approximately 44,300; results were sorted by relevance and the first 50 were screened, consistent with common practice for managing this source.

A7. Cochrane Library

Search string:
background music AND (concentration OR attention OR cognitive performance OR memory OR learning OR motivation)

Date searched: June 2026. *Records returned:* 29 (screened in full); no records met the eligibility criteria.

A8. Manual search: Educational Research Review

Method: in-journal search for “music” via ScienceDirect.
Date searched: March 2026. The manual search returned 3 records, none of which were incorporated into the charted set (one already identified via database searching, the others not taken forward within the project timeframe).

A9. Backward citation searching

Method: the reference lists of included studies were screened for further eligible records, and additional targeted searches were conducted to verify and locate the full citation details of specific studies once identified.
Date completed: 7 July 2026. *Records identified:* 4, contributing to the “additional records identified through other sources” count in Figure 3.1.

Appendix B: Declaration of the use of artificial intelligence (AI)

During the preparation of this thesis, I used Claude (Anthropic) as a language- and consistency-editing aid. Specifically, the tool was used to check spelling and grammar, to suggest improvements to the clarity and flow of individual sentences, and to check the internal consistency of the manuscript, for example the agreement between the text, the tables, and the reference list. I reviewed every suggestion and accepted, rejected, or rewrote it in my own words. The research question, the literature search, study selection, data charting, the classification of studies, and all analysis and interpretation are my own work. I have located, read, and verified every source cited in this thesis, and I take full responsibility for its final content.